The Incompleteness of Observation

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Why Physics' Biggest Contradiction Might Not Be a Contradiction at All

With Complete Mathematical Detail

Alex Maybaum — March 2026

The Worst Prediction in Physics

Physics has two spectacularly successful theories. Quantum mechanics describes the behavior of atoms, particles, and light. General relativity describes gravity, space, and time. Each has been confirmed to extraordinary precision. They have never disagreed with any experiment.

They disagree with each other.

Ask quantum mechanics how much energy empty space contains and it gives you a staggering number: roughly 10^{113} joules per cubic meter. Ask general relativity the same question — read the answer off the expansion rate of the universe — and you get about 6×10^{-10} joules per cubic meter. The ratio is 10^{122}. For scale, the number of atoms in the observable universe is about 10^{80}. This is not a close call.

For decades, the assumption has been that something is deeply broken — that one or both calculations contain an error, and that finding the mistake will point the way to a unified theory of everything.

This paper argues the opposite. Neither calculation is wrong. On the framework's reading they disagree because they *must*, and that massive 10^{122} discrepancy isn't a failure at all: it is a sign that quantum mechanics is not the fundamental bedrock of reality, but an emergent description forced upon us by our limited vantage point. Whether gravity in fact ignores the emergent zero-point energy is an open step in how gravity couples to the two descriptions.

The argument is built from a chain of mathematical proofs, each feeding into the next. This document explains what the paper claims, walks through the logic of every major proof, and shows how they connect.

The Starting Point: Observation Exists

Every mathematical proof starts from assumptions, and this framework has exactly one. It doesn't mention quantum mechanics. It doesn't mention general relativity. It is:

Observation occurs.

An observer records distinguishable outcomes of interactions with a system not wholly under the observer's control. This is the cogito of Descartes — "I think, therefore I am" — made mathematically precise. It is the one empirical fact that cannot be doubted: if you are reading this sentence, observation is occurring.

The paper formalizes this as a definition:

Definition. An *observation* is a triple (S, φ, V): a total system S, a deterministic dynamics $\varphi : S \to S$, and an observer $V \subsetneq S$ — a proper subsystem with finitely many distinguishable internal states, coupled to the complement $H = S \setminus V$ through φ.

This single sentence contains no physics. It is weaker than classical mechanics (no continuity, no Hamiltonian, no Lagrangian). A shuffled deck of cards satisfies it. A finite cellular automaton satisfies it. Any finite computation satisfies it. The paper shows that three structural lemmas follow from this definition alone:

A note on "exactly one." The framework's foundational work has since looked harder at what "observation occurs" actually assumes, and found it is most honestly stated as **two** axioms rather than one. The first — that a distinction occurs, that there is some this-rather-than-that — genuinely cannot be doubted. The second — that the distinction *recurs*, that there is more than one moment — is a real further assumption: a single frozen instant, however full of difference, contains no time, and time cannot be squeezed out of difference alone. The framework assumes both, and a no-go result shows the second does not follow from the first. This is a strength rather than a hedge: the foundation can point to exactly which single ingredient is assumed rather than indubitable. A further question is genuinely open: whether the observer's status as a *proper part* of a larger whole — its embeddedness — is itself a third such assumption, or instead follows from the first axiom. This turns on a subtle point about what "a distinction occurs" means, it is under review, and — importantly — it changes no prediction the framework makes: it bears only on how the foundation's assumptions are counted and described. The companion paper *Physics Modulo Gauge* develops the foundational structure in full, as its Part I; for the purposes of this overview, "observation occurs" can be read as that two-part (possibly three-part) foundation, and nothing below depends on the refinement.

Lemma 1 (Finiteness). *The observer has finitely many distinguishable internal states, so the visible configuration space $\mathcal{C}_V$ is finite, with a discreteness scale ϵ providing a finite minimal cell volume.* There's a smallest meaningful size ε — you can't resolve anything smaller. This means the configuration space is finite,

not continuous. This matters because finite systems have a property infinite systems don't — they must eventually return to their starting state (Poincaré recurrence). In Part I, ε is left unspecified. In Part II, self-consistency forces $\varepsilon = 2\sqrt{\sigma}$ l_p, where σ is the entropy carried by one boundary mode — twice the Planck length at $\sigma = 1$, one nat per boundary mode.

Lemma 2 (Causal partition). *An observer is a proper subsystem $V \subsetneq S$. The complement $H = S \setminus V$ is the hidden sector.* The total phase space splits into two pieces:

$$\Gamma = \Gamma_V \times \Gamma_H$$

Γ_V is the visible sector (what the observer can access). Γ_H is the hidden sector (what they cannot). The total Hamiltonian splits correspondingly:

$$H_{\text{tot}} = H_V + H_H + H_{\text{int}}$$

H_V governs the visible sector alone. H_H governs the hidden sector alone. H_int couples them — it's how the two sectors talk to each other. Without H_int, the two sectors would evolve independently and the observer would never feel the hidden sector's influence.

Lemma 3 (Selected measure). *The counting measure on S — assigning equal weight to each state — is selected among the measures invariant under φ as the one of maximal entropy; on a single orbit it is the only invariant measure.* The observer uses standard Kolmogorov probability theory. No exotic probability theories, no negative probabilities, no quantum probability — just ordinary probability. This is what makes the result surprising: we're putting in classical probability and getting out quantum mechanics.

That's it. The precise claim is layered. At the finite observable-law level, the resulting stochastic law has an exact fixed-basis Hilbert/unitary/Born-form representation ($S \iff D \iff Q_{\text{fb}}$); when C4 readback holds, the OI sector additionally carries genuine predictive memory. The Schrödinger/unitary interpolation and the $p = |U|^2$ dictionary belong to that exact representation layer. What does **not** follow from the definition alone is that every coherent preparation, noncommuting instrument, and composite is already one standard quantum operational theory; on a finite carrier it follows exactly when five further operational conditions hold — valid probabilities, trivial-ancilla consistency, inert spectators, full reversible control, and iterated composition — and each is genuinely additional to embedded observation (Main §3.4; GR §3.3). Bell is a separate fork: a response-complete deterministic completion that is both measurement-independent and ontically parameter-independent is Bell-local, so the framework retains measurement independence and takes ontic parameter dependence for Bell-violating substrata while preserving operational no-signaling. The hidden-sector conditions are:

C1: Non-zero coupling (H_int $\neq$ 0). The visible and hidden sectors interact. Information flows between them. Without this, the observer's room is perfectly isolated — nothing interesting happens.

C2: Memory persistence (τ_S $\ll$ τ_B). The hidden sector's *coarse-grained mixing time* — the rate at which a stored record of visible history degrades — is long compared with the accessible window. This is persistence, not slowness: fast hidden dynamics that conserve the record serve equally well, and a shift-register bath rotating five cells per visible step supports a maximal readback gap ([Main §1.3]). τ_S is the timescale of visible-sector processes; τ_B is the timescale of hidden-sector processes. This is the *opposite* of the usual assumption in physics. Normally, people assume the environment is fast and chaotic (a "heat bath" that quickly forgets everything). Here, the environment is slow and has a long memory. This is what makes the dynamics non-Markovian.

C3: Sufficient capacity ($\log_2$|C_H| $\geq$ I*). The hidden sector has enough distinguishable states to encode the conditional past-future information the observer actually sees — the per-process bound, not a crude "many more degrees of freedom" comparison. It follows from readback by data processing in any faithful realization, and framework realizations saturate it ([Main §1.3]).

The definition sets the stage. The conditions determine what kind of show plays on it. The next section explains why the cosmological horizon satisfies all three.

The Observer's Blind Spot

Light travels at a finite speed, and the universe has a finite age. Put those together and every observer has a horizon — a boundary beyond which no signal has had time to arrive. Everything beyond that boundary is ordinary physics: fields, particles, radiation. But it is structurally inaccessible. Not because our telescopes aren't good enough, but because the geometry of spacetime forbids it. No technology that obeys the speed of light can reach past it.

At an observation event, direct signals come from its causal past. A physical event horizon can impose a lasting direct-access restriction under the spacetime model's hypotheses. This is distinct from inference: correlations in retained records may determine some variables that are never read directly. The abstract fact that an observer sees a proper part of a system does not prove that its full history leaves every hidden state ambiguous.

For a known deterministic model, the reduced prediction averages over the hidden posterior conditioned on the observer's available records. That posterior can narrow as records accumulate and can become a point mass. If several compatible hidden states still predict different outcomes, the reduced prediction is probabilistic; if the records identify the state, the next outcome is fixed within that model. The swap $(v, h) \mapsto (h, v)$ is the simplest example: (X_0, X_1)

identifies both initial bits, although $X_2 = X_0$ makes prediction depend on history. The precise criterion uses record fibres [Substratum Proposition 24.5], with record retention and allowed interventions stated separately.

Concretely, several hidden states may be compatible with the present visible state x. Their next visible outputs can differ, and their conditional weights define a transition law. Those weights must use the retained history, rather than assuming the initial hidden prior remains unchanged after observation. Averaging over a preparation ensemble does not prove that an individual state is unknowable after conditioning. This is a stochastic reduced description, including deterministic conditional predictions as a special case.

But the hidden sector beyond the cosmological horizon is not like that. It differs in three specific ways, and the paper proves that their conjunction changes everything.

It's coupled. The horizon is not a static wall. Stress-energy conservation enforces continuous dynamical correlations across it. Matter crosses the horizon, and the horizon area adjusts in response to interior energy density. Information flows in both directions. (Condition C1.)

It's slow. The hidden sector's correlation time is set by the Hubble timescale — roughly 10^{17} seconds, the age of the universe. Any laboratory experiment operates on timescales of 10^{-15} seconds or shorter. The ratio is 10^{-32}. The hidden sector cannot "reset" between your measurements. Every correlation it picks up from one experiment is still there when the next one begins. This is the *opposite* of the standard Markovian regime, where the environment decorrelates fast. Here, it never decorrelates at all. (Condition C2.)

It's vast. The hidden sector has roughly 10^{122} independent degrees of freedom at one nat each — the Bekenstein-Hawking entropy of the cosmological horizon. No experiment you could ever perform would appreciably disturb its state. Its memory never saturates. (Condition C3.)

A fast environment with vast capacity would wash out correlations (Markovian noise). A slow environment with limited capacity would eventually fill up and stop recording. What the observer needs is an environment that couples to it, retains the record long enough to be read, has the capacity to hold it, and feeds it back — and it is the feedback (C4) that does the work, the other three following from it or supporting it. Such an environment sustains persistent conditional memory — transition probabilities that depend on the history, not just the present. A distinct property, which the paper calls *P-indivisibility* — the system's transition probabilities at different times cannot be broken into independent steps — holds across the recurrence cycle, and on accessible windows only where a divisibility witness shows it.

Partition-Relativity (§1.4)

This is the first real proof in the paper, and it's beautifully simple.

What it proves: Under the canonical measure of Lemma 3, the emergent description (what the observer sees) depends *only* on the partition — on which degrees of freedom are visible and which are hidden. For a general hidden prior the measure is a third input ([Main §1.4]).

The formula:

$$T_{ij}(t_2, t_1) = \int_{\Gamma_H} \delta_{x_j}[\pi_V(\phi_{t_2-t_1}(x_i, h))]\, d\mu(h)$$

Unpacking each symbol:

- **T__ij**: The probability of transitioning from visible state x_i to visible state x_j in the time interval from t_1 to t_2. This is what the observer measures.
- **(x__i, h)**: The complete state — visible part x_i, hidden part h.
- **φ__{t2-t1}**: The deterministic evolution. Takes the complete state at time t_1 and returns the complete state at time t_2. Uniquely determined by the definition.
- **π__V**: Projection onto the visible sector. Takes a complete state (x, h) and returns just x.
- **δ__{xj}[...]**: The Kronecker delta. Equals 1 if the visible part ended up at x_j, equals 0 otherwise.
- **dμ(h)**: Integration over all possible hidden states, weighted by the hidden-sector measure μ — canonically the Liouville measure, the continuum form of Lemma 3's counting measure.

In plain English: For each possible hidden state h, check whether starting at (x_i, h) and evolving forward lands the visible part on x_j. Count up all the hidden states where this happens, weighted by how likely each hidden state is. The result is the probability of the transition x_i → x_j.

The proof: The formula has exactly three inputs: (1) the dynamics φ_t — fixed by the definition, (2) the partition (Γ_V, Γ_H) and projection π_V — fixed by Lemma 2, and (3) the measure μ. Lemma 3 supplies a canonical measure for input 3 by a selection principle: the counting measure, the maximal-entropy invariant measure, whose continuum form is the Liouville measure. With input 1 fixed by the definition and input 3 set to the canonical measure, the only free input is the partition. Therefore: under the canonical measure, everything about the emergent description depends only on the partition. QED.

Where the Liouville measure comes from: The observer needs a "prior" — a way to weight the hidden states. Invariance under the dynamics does not select one. On a single orbit the invariant measure is unique and uniform, but

every invariant set of the bijection carries its own invariant measure, so a bijection with two or more cycles has a whole family of them; in the continuum, Liouville measure is invariant but is not, in general, the unique absolutely continuous invariant measure. The counting measure is selected among the invariant measures as the one of maximal entropy — a selection principle, not uniqueness from invariance alone. An embedded observer with incomplete knowledge of the total energy averages over energies with a smooth prior, recovering the Liouville measure on a phase-space band — the continuum form of that maximal-entropy selection. Singular invariant measures concentrated on atypical short orbits correspond to observer-lacking substrata and are outside scope by Lemma 3. The canonical measure is the default, not a constraint on representations: a realization may carry its own fixed hidden prior μ_H, the emergent law is then a function of (φ, partition, μ_H), and the same bijection and partition under different priors generally give different visible laws ([Main §1.2, Lemma 3 and its remarks]).

Emergent Stochasticity and the Slow-Bath Regime (§§2.1–2.2)

The total system is deterministic. If you knew both x and h, you'd know the future with certainty. But the observer knows only x. Different hidden states h send the same visible state x to different futures.

Example: visible state is "Heads." Hidden state could be any die value 1–6. If the die is 1 or 2, the dynamics flip the coin to Tails. If the die is 3–6, the coin stays at Heads. The observer doesn't know the die, so they see: P(Heads $\rightarrow$ Tails) = 2/6 = 1/3. The randomness is epistemic (from ignorance) not ontological (from fundamental indeterminacy).

In a normal "heat bath" scenario, the environment is fast and chaotic. It scrambles any information you write into it before you can read it back. This produces Markovian (memoryless) dynamics — each step is independent of previous steps.

C2 inverts this — the condition itself is persistence, the record surviving to readback, with a slow hidden sector our universe's comfortable route (a fast bath that conserves and routes the record serves equally). The hidden sector here is slow. When the visible sector interacts with it (writing information through H_int), the information stays there. At the next interaction, the hidden sector reads back what was written before. The observer sees history-dependent transition probabilities — what happens next depends on what happened before.

This is non-Markovian dynamics. It's the key ingredient that separates quantum mechanics from classical stochastic processes.

The P-Indivisibility Theorem (§2.3)

What it claims. If a reversible (bijective) deterministic system on a finite state space is split into a visible and a hidden sector, and the one-step visible transition matrix, computed with the hidden sector uniformly distributed, is not a permutation, then the visible process is P-indivisible across the recurrence cycle: recurrence returns the visible map to the identity, and no valid transition matrix carries the early map to it. This is a recurrence-scale statement. On accessible windows the proved statement is the accessible-timescale lemma below: a (C4) gap forces positive conditional memory, which does not by itself make the process P-indivisible there.

What "P-indivisible" means. A stochastic process is "P-divisible" if you can always find a valid transition matrix connecting any two time points. Mathematically: for any times t_1 < t_2 < t_3, there exists a stochastic matrix Λ such that:

$$T(t_3, t_1) = \Lambda(t_3, t_2) \cdot T(t_2, t_1)$$

where Λ has non-negative entries and rows summing to 1. "P-indivisible" means this fails — the "intermediate propagator" would need negative entries, which means it's not a valid probability matrix.

Breuer, Laine, and Piilo introduced "information backflow" — the system's distinguishability *increasing* over time — as the signature of non-Markovian dynamics. In a classical Markov process, you can only lose information (mixing); the same holds in any P-divisible process, so backflow certifies P-indivisibility, though a P-indivisible process need not show it. Where it does, information comes back. This is exactly what quantum systems do — interference, revivals, and non-classical correlations all involve information returning from where it was stored.

The setup. We work on finite sets (Lemma 1). The visible sector has states C_V = {x_1, x_2, …} with |C_V| ≥ 2. The hidden sector has states C_H = {h_1, h_2, …}. The total dynamics is a bijection φ on C_V × C_H. The transition matrix is:

$$T_{ij} = \frac{|\{h \in \mathcal{C}_H : \pi_V(\varphi(x_i, h)) = x_j\}|}{|\mathcal{C}_H|}$$

The key tool — total variation distance:

$$d(p, q) = \frac{1}{2} \sum_k |p_k - q_k|$$

This measures how distinguishable two probability distributions are. If d = 1, they're perfectly distinguishable. If d = 0, they're identical. For P-divisible processes, d can only decrease or stay constant.

Step 1 — Recurrence. φ is a bijection on a finite set. Keep applying φ and you must eventually return to where you started — there are only finitely many states to visit. Formally: there exists N such that $\varphi\hat{}N = \mathrm{id}$. So $T\hat{}(N) = I$, and:

$$d(\delta_i T^{(N)}, \delta_j T^{(N)}) = d(\delta_i, \delta_j) = 1$$

After N steps, states that started distinguishable are still perfectly distinguishable.

Step 2 — Strict contraction. Suppose T is not a permutation matrix. (This does not follow from C1, which asserts only non-zero coupling; it is the separate non-permutation witness the theorem assumes at this step.) So there exist states i, j, l where both T_il > 0 and T_jl > 0. The total variation distance after one step:

$$d(\delta_i T, \delta_j T) = \frac{1}{2} \sum_k |T_{ik} - T_{jk}| < 1$$

The inequality is strict because the distributions overlap. Distinguishability has decreased.

Step 3 — The punchline. At t = 1: d < 1 (distinguishability decreased). At t = N: d = 1 (distinguishability restored). The distinguishability went down then came back up — non-monotonic behavior. A P-divisible process can only have non-increasing distinguishability. Therefore the process is P-indivisible. QED.

The proof uses almost nothing — just that the dynamics is a bijection on a finite set (the definition and Lemma 1) and that the coupling is non-trivial (C1). It is purely combinatorial.

The Accessible-Timescale Lemma (§2.3 continued)

The recurrence proof shows P-indivisibility exists as a mathematical property. But the recurrence time is absurdly long — for the cosmological case, it's $e^{(10}122)$ years. Nobody will ever observe it.

The accessible-timescale lemma shows that conditional memory is present on *laboratory timescales*, independently of recurrence. A revival of distinguishability on those timescales — information backflow — is a further property that readback alone does not supply.

The mechanism: At each interaction (timescale τ_S), the coupling H_int transfers some information from the visible sector to the hidden sector. Call the amount I_0. Between interactions, the hidden sector's correlations decay with a rate set by its spectral gap $\Delta \sim 1/\tau$_B.

The decay per visible-sector step is:

$$e^{-\Delta\tau_S} \approx 1 - \frac{\tau_S}{\tau_B}$$

When τ_S $\ll \tau$_B (C2), this is very close to 1 — almost no decay. The hidden sector remembers almost perfectly between steps. After k steps, the cumulative decay is:

$$e^{-k\Delta\tau_S} \approx 1 - \frac{k\tau_S}{\tau_B}$$

As long as k $\cdot \tau$_S $\ll \tau$_B, the hidden sector retains ~k bits of visible-sector history. The mutual information satisfies:

$$I(X_{<t}; X_{>t} \mid X_t) \geq I_0 \left(1 - \frac{k\tau_S}{\tau_B}\right)$$

For the cosmological case: τ_S ~ 10^{-15} s, τ_B ~ 10^{17} s. Even after k = 10^{20} steps, k $\cdot \tau$_S/τ_B ~ 10^{-12} — negligible. The hidden sector remembers everything.

The role of C3: The hidden sector's memory capacity is log_2(|C_H|) bits. If k bits of history are written but the capacity is only m < k bits, old data gets overwritten. C3 ensures m is large enough that the memory never saturates on observable timescales.

The Coin-and-Die Model (§2.4)

The paper builds a concrete toy model to make the mechanism tangible.

Setup: - Visible: x $\in$ {0, 1} (a coin: 0 = Heads, 1 = Tails) - Hidden: h $\in$ {1, 2, 3, 4, 5, 6} (a die) - Total: 12 states

The permutation σ:

Input state	Output state	What happens
(0, 1)	(1, 1)	Coin flips, die stays
(1, 1)	(0, 1)	Coin flips, die stays
(0, 2)	(1, 2)	Coin flips, die stays

Input state	Output state	What happens
(1, 2)	(0, 2)	Coin flips, die stays
(0, 3)	(0, 4)	Coin stays, die changes
(0, 4)	(0, 3)	Coin stays, die changes
(0, 5)	(0, 6)	Coin stays, die changes
(0, 6)	(0, 5)	Coin stays, die changes
(1, 3)	(1, 4)	Coin stays, die changes
(1, 4)	(1, 3)	Coin stays, die changes
(1, 5)	(1, 6)	Coin stays, die changes
(1, 6)	(1, 5)	Coin stays, die changes

Every swap is a transposition (a $\leftrightarrow$ b), so $\sigma^2 = \text{id}$ (apply twice, everything returns).

Checking the conditions: C1 (coupling): die values 1 and 2 flip the coin ✓. C2 (memory persistence): $\sigma^2 = \text{id}$ means recurrence time is 2 steps, giving $\tau_S/\tau_B = 1/2$ ✓. C3 (sufficient capacity): 6 hidden states vs 2 visible states ✓.

Computing T(1,0). Start at x = 0 (Heads). All 6 die values are equally likely.

- h = 1: $\sigma(0,1) = (1,1) \rightarrow$ Tails
- h = 2: $\sigma(0,2) = (1,2) \rightarrow$ Tails
- h = 3: $\sigma(0,3) = (0,4) \rightarrow$ Heads
- h = 4: $\sigma(0,4) = (0,3) \rightarrow$ Heads
- h = 5: $\sigma(0,5) = (0,6) \rightarrow$ Heads
- h = 6: $\sigma(0,6) = (0,5) \rightarrow$ Heads

$P(0 \rightarrow 0) = 4/6 = 2/3$, $P(0 \rightarrow 1) = 2/6 = 1/3$. By the same logic for x = 1:

$$T(1,0) = \begin{pmatrix} 2/3 & 1/3 \\ 1/3 & 2/3 \end{pmatrix}$$

Distinguishability at t = 1:

$$d(\delta_0 T, \delta_1 T) = \frac{1}{2}(|2/3 - 1/3| + |1/3 - 2/3|) = 1/3$$

Started at d = 1. Now d = 1/3. Distinguishability decreased.

What Markov would predict at t = 2: Apply the same transition matrix again:

$$T(1,0)^2 = \begin{pmatrix} 5/9 & 4/9 \\ 4/9 & 5/9 \end{pmatrix}$$

Distinguishability would drop to d = 1/9. More mixing.

What actually happens at t = 2: $\sigma^2 = \mathrm{id}$. Every state returns to its starting point.

$$T(2,0) = I = \begin{pmatrix} 1 & 0 \\ 0 & 1 \end{pmatrix}$$

Distinguishability is back to d = 1. Complete un-mixing. Impossible for a Markov process.

The smoking gun — negative entries. If there were a valid stochastic matrix $\Lambda(2,1)$ connecting steps 1 and 2:

$$\Lambda(2,1) = T(2,0) \cdot [T(1,0)]^{-1} = I \cdot \begin{pmatrix} 2 & -1 \\ -1 & 2 \end{pmatrix} = \begin{pmatrix} 2 & -1 \\ -1 & 2 \end{pmatrix}$$

The entries -1 are negative. No valid stochastic matrix exists. **This is P-indivisibility.**

The mechanism in detail. The die works as a memory register. At step 1, if the coin was at 0 and the die was at 1, the coin flips to 1 but the die stays at 1. The die value 1 now encodes the information "the coin was at 0 and I flipped it." At step 2, σ sees (1, 1) and flips it back to (0, 1). The die read its own memory and reversed the flip. C1 (coupling) allows writing to the memory. C2 (memory persistence) ensures it isn't erased between reads. C3 (sufficient capacity) ensures there's enough room. C4 (history readback): at step 2 the unchanged die register controls the reverse flip — the dynamics reads back the record written at step 1. In this toy, C1 through C4 explain the memory mechanism — the history readback; the explicit negative-propagator calculation above independently establishes the stronger P-indivisibility witness, and the process is thereby, via the stochastic-quantum correspondence, the carrier of a nontrivial quantum representation in compressed form under (T).

Why conditions C2 and C3 matter physically

The P-indivisibility theorem needs only a non-permutation rooted map and finiteness (with the uniform hidden prior, any non-trivial coupling (C1) supplies one). So why does the paper insist on slow memory (C2) and vast capacity (C3)?

Because recurrence-scale P-indivisibility shows up only at absurd timescales. On accessible timescales the relevant property is conditional memory: C2 ensures the memory persists on timescales accessible to actual experiments, not just at cosmic recurrence times. C3 ensures the hidden sector never runs out of room to store information — if it saturates, later imprints overwrite earlier

ones, and the process becomes effectively memoryless. Together with C4's readback — the load-bearing condition (`review4_probes.py`: storage and capacity without readback do not suffice) — C2 and C3 make the accessible memory strong, persistent, and observationally relevant; the conditional memory M_t is the diagnostic that readback makes measurable.

The Stochastic-Quantum Correspondence (§3.1 and Appendix A)

This is the key link. Section 2 proved that the embedded observer's dynamics are P-indivisible across the recurrence cycle. Section 3 shows how the correspondence supplies the quantum representation for exactly this sector — admitted by every finite law, nontrivial here.

The core statement. Any P-indivisible stochastic process on a finite configuration space of size n can be embedded into a unitarily evolving quantum system. Specifically, there exists a Hilbert space H (dimension $\leq n^3$) and a unitary operator U(t) such that:

$$T_{ij}(t) = \sum_{\alpha} |\tilde{U}_{(i,0)\,(j,\alpha)}(t)|^2 \qquad \text{(ancilla-marginal; the bare visible form is the } m_a = 1 \text{ special case)}$$

This is the Born rule. The left side is the transition probability computed by averaging over hidden states (the classical formula from partition-relativity). The right side is the quantum mechanical probability — the squared modulus of a matrix element of the unitary evolution operator. The equivalence is not approximate. It is not an analogy. It is a mathematical identity.

Two routes to the same layer — and the in-house one is now primary. The imported route uses Barandes' stochastic-quantum correspondence (2023–2025), which supplies the compressed form: P-indivisibility means transition probabilities can't be factored through intermediate times — try it and you get "negative probabilities." In quantum mechanics, this is *exactly what happens*: probability amplitudes combine to produce interference patterns that don't factorize classically. What Barandes proved is that every finite-configuration process in his class — a broad one, including Markov chains — can be regarded as a subsystem of a unistochastic process, so the two descriptions are notational variants of one object. The breadth matters: the representation's existence is not the framework's distinctive result, and what P-indivisibility adds is that the dynamics so represented is genuinely indivisible.

The in-house route, given in Appendix A, uses Stinespring's dilation theorem (1955), and it is now primary — its permutation unitary is exactly the object behind the equivalence theorem's internal $D \Rightarrow Q_{\text{fb}}$ leg ([Main §3.4]): a deterministic bijection on a finite product space defines a permutation unitary;

tracing out the hidden sector with the Liouville measure produces a completely positive quantum channel whose diagonal elements recover the classical transition probabilities exactly. This second route requires only textbook results. Both reach the transition-statistics layer and neither reaches the operational instrument algebra, which stays open; together they ensure that *layer* rests on no single recent result.

Where the quantum features come from:

- **The Schrödinger equation** arises because U(t) is differentiable. Any continuous one-parameter *group* of unitaries — U(s+t) = U(s)U(t) — can be written as U(t) = exp(-iHt/$\hbar$) for one fixed Hermitian H (Stone's theorem); the framework's realized processes supply exactly such a group at horizon times (the fixed-$\hat{\mathrm{H}}$ lemma), while a general smooth family without the group property would need a time-dependent generator. Feynman famously remarked of this equation in Volume III, Lecture 16: "Where did we get that equation from? Nowhere. It is not possible to derive it from anything you know. It came out of the mind of Schrödinger." The derivation above is the answer that wasn't available in 1963. The equation is not "from nowhere" — it is the differential form of unitary evolution, and unitary evolution is available for embedded observers' processes, internally through the permutation unitary of any finite reversible realization and in compressed form under (T) ([Main §3.4]) — nontrivially so where the dynamics is P-indivisible, which on a fixed finite reversible representative holds across its recurrence cycle whenever a rooted visible map is not a permutation.

- **The Born-form dictionary** is the ancilla-marginal rule $T_{ij} = \sum_\alpha |\widetilde{U}_{(i,0)(j,\alpha)}|^2$ in general, reducing to $T_{ij} = |U_{ij}|^2$ only in the trivial-ancilla case. Inside Q_{fb} that quadratic readout is part of the definition of the equivalent representation. The permutation-unitary construction does not, however, select the exponent: its dilated matrix entries are 0 or 1, so replacing the square by $|\widetilde{U}|^p$ leaves the same marginal table for every $p > 0$. Why the quadratic functional is the physical dictionary is therefore a question about the dictionary's origin, not a gap in the fixed-basis equivalence.

- **The action scale** $\hbar$ enters when converting from the dimensionless unitary to a dimensionful Hamiltonian: $\hat{\mathrm{H}} = \mathrm{i}\hbar\ (\partial \mathrm{U}/\partial \mathrm{t})\ \mathrm{U}\dagger$. The value of $\hbar$ cannot be determined from the dimensionless transition data alone — it requires additional physical input from the partition geometry (§5).

- **Bell correlations.** Bare non-factorization is not enough to select the quantum correlation set. The stochastic causal-local indivisible construction has a Tsirelson bound, but the framework's response-complete deterministic layer is more restrictive: pointwise graph locality plus a measurement-independent pre-setting ensemble gives a Bell-local model with $|S_{\mathrm{CHSH}}| \leq 2$. The framework therefore takes the ontic-parameter-

dependent branch for Bell-violating substrata while retaining operational no-signaling (Main §3.3). Unrestricted/global composites remain a post-quantum negative control, so neither hidden memory nor indivisibility is a selector of the quantum Bell set.

The Phase-Locking Lemma (§3.1 continued)

A potential objection: the relation T_ij = |U_ij|^2 throws away phase information. Different unitaries could give the same transition probabilities. Does this make the quantum description ambiguous?

The complete two-branch answer requires the sign-aware classification hypothesis H-BG ([GR §3.3]); Fourier extraction alone does not classify the unlabelled mode correspondences.

Setup: The transition probability at time t is:

$$T_{ij}(t) = \left| \sum_k V_{ik}\, e^{-iE_k t}\, V_{jk}^* \right|^2$$

where V_ik = ⟨i|k⟩ are the overlaps between the configuration basis and the energy eigenbasis, and E_k are the energy eigenvalues. Expanding the square:

$$T_{ij}(t) = \sum_{k,l} V_{ik}\, V_{jk}^*\, V_{jl}\, V_{il}^*\, e^{-i(E_k - E_l)t}$$

The Fourier trick: Once a mode correspondence has been fixed, this is a sum of oscillating terms at frequencies ω_kl = E_k - E_l. If all these frequencies are distinct (condition G2: non-degenerate energy gaps), you can extract each coefficient by Fourier transform:

$$a_{ij}^{kl} = V_{ik}\, V_{jk}^*\, V_{jl}\, V_{il}^*$$

Extracting the moduli: Setting i = j: $a_{ii}^{kl} = |V_{ik}|^2 |V_{il}|^2$. If none of the overlaps are zero (condition G3), all moduli |V_ik| are determined.

Extracting the phases: Write V_ik = |V_ik| e^{iφ_ik}. The argument of the Fourier coefficient gives:

$$\arg(a_{ij}^{kl}) = (\varphi_{ik} - \varphi_{il}) - (\varphi_{jk} - \varphi_{jl})$$

The only transformation preserving all double differences is φ_ik → φ_ik + α_i + β_k — just relabeling (choosing a different phase convention for the basis

states). Once you fix these conventions, all remaining phases are determined up to one further discrete ambiguity: the complex-conjugate Hamiltonian $H \to -H^*$, an antiunitary twin producing identical transition probabilities at every time.

Conditional conclusion: Under H-BG and the gap/overlap hypotheses on both Hamiltonians, continuous-time transition data determine the Hamiltonian up to an energy shift, diagonal rephasing and the antiunitary conjugate $H \to -H^*$. The last ambiguity is a global orientation alternative. General ancilla-case completeness remains open beyond the tested generic cases.

Bell Inequality Violations (§3.2)

This is the question everyone asks: isn't this ruled out by Bell's theorem?

What Bell's theorem actually requires. For the response-complete deterministic layer, the decisive conjunction is measurement independence plus ontic parameter independence. If one pre-setting microstate fixes the responses at both wings, one common setting-independent ensemble is used, and the Bell-relevant coupling range does not connect the wings before readout, the result is an ordinary local hidden-variable model and $|S_{\mathrm{CHSH}}| \leq 2$. Hidden memory and indivisibility do not change that conclusion.

The branch the framework takes. The framework declines the measurement-dependent/superdeterministic horn and instead relinquishes **ontic parameter independence**. In the specific SM/GR representative this need not be a static all-to-all sector: [SM §3.1] already makes the coupling graph state-dependent, so the branch-(a) target is preparation-indexed adjacency — a geometric bounded-range reference graph in the uniform/vacuum class, with bounded-degree long links (or an equivalent nonlocal response map) supplied by an entangling preparation. The operational distribution remains no-signaling. The unresolved cost is H-Bell's curvature-stability condition: the prepared graph must retain the metric/Ollivier–Ricci continuum behavior required by the Einstein reconstruction. That condition is now downstream of a harder one. The cited convergence theorem is arbitrary-dimensional only for the manifold-distance-weighted graph metric, and for the degree-6 cubic reference graph the intrinsic hop metric is exactly ℓ_1 — a $\sqrt{2}$ diagonal stretch that does not decay at any scale — so the hop-metric curvature route FAILS for the reference family, before any Bell edge is added, and no $D = 3$ extension of that theorem would rescue it. The emergent geometry itself is not in question: the substratum's dispersion relation is isotropic at leading order. What must be re-founded is the curvature functional, on the propagation/Laplacian geometry rather than on shortest-path counts [SM §3.2]. The measurement-dependent alternative is retained only as a quantified counterfactual. Main §3.3 proves the model-independent necessary bound $I_{\mathrm{ont}} \geq 0.0309409\ldots$ bits for the quantum value; this is not an achievable cost. Hall–Branciard's optimized causal CHSH

construction in its model class uses approximately 0.080 bits, whereas the approximately 0.046-bit optimum is retrocausal; OI's one-way branch does not adopt the latter.

The stochastic Tsirelson result is a different layer. Barandes–Hasan–Kagan bound a weaker class of causally local indivisible *stochastic* processes by $2\sqrt{2}$. Their causal-independence condition is weaker than Bell factorization, so that theorem does not turn the response-complete deterministic completion into a local quantum hidden-variable model. Jarrett/Fine diagnostics remain useful; at the deterministic layer the Bell mechanism is ontic parameter dependence, while stochastic causal locality belongs to the weaker reduced-law description.

The Characterization Theorem (§3.3)

It's not enough to show that embedded observation *produces* QM (sufficiency). The paper shows the observed non-Markovian dynamics *require* embedded observation under C1–C4 (necessity, per condition; the C2 leg conditional on the mixing hypothesis in its slow-bath form). The full logical chain:

- Barandes proved: an exact correspondence between quantum systems and the source's indivisible class, defined at the level of first-order conditionals (the framework's link to it is the translation hypothesis (T), [Main §3.1])
- Section 2.3 proved: a non-permutation coupling $\Longrightarrow$ P-indivisibility at the recurrence scale, and C4's readback $\Longrightarrow$ accessible non-Markovianity, quantitatively (sufficiency). The two conclusions are different notions and neither is generic quantum non-Markovianity: accessible non-Markovianity is finite-horizon memory in the visible law, while P-indivisibility is failure of divisibility of the visible population law and needs the recurrence and non-permutation hypotheses, not C1/C3/C4 alone
- Section 3.3 proves: accessible-window P-indivisibility under a full-support visible prior $\Longrightarrow$ C1, C3, C4 (necessity); C2's slow form within the mixing class (the physical-memory theorem, slow-bath form)
- Combined: **accessible non-Markovianity $\Longleftrightarrow$ embedded observation under C1, C3, C4 (for some accessible order k, on each finite accessible horizon $K \geq k+1$, with the readback at k) — C2 is not needed for the equivalence and carries its physical content separately, through the physical-memory theorem (persistence form without a mixing hypothesis, slow-bath form conditional on it) — with a fixed-basis unitary quantum representation internal and universal ([Main §3.4]), the compressed form supplied across the correspondence's class under (T), nontrivially where the dynamics is P-indivisible**

Necessity of C1 (coupling). With no hidden influence at any accessible step, every visible transition is a fixed function of the visible present — the process is Markov, hence not the observed one. A non-permutation one-step matrix is

the cheapest *witness* of C1, sufficient but not necessary: influence can hide at step one and surface later (the delayed-revival family).

Necessity of C2 (memory persistence), the slow form conditional on mixing. Between coupling events (separated by τ_S), the hidden sector evolves under its own Hamiltonian. The convergence to equilibrium is:

$$\|e^{\mathcal{L}_H \tau_S} \mu_H(\cdot|x_i) - \mu_{\text{eq}}\|_{\text{TV}} \leq C\, e^{-\Delta \tau_S}$$

In the fast-bath regime ($\Delta \cdot \tau$_S $\gg 1$), this is exponentially small. The hidden sector forgets everything between interactions. Each transition is computed against the same equilibrium distribution, so T^(k) = T^k — a Markov chain, hence P-divisible. Contrapositive, within the mixing class: P-indivisibility requires τ_S $\ll \tau$_B — conditional because the convergence bound presupposes a relaxation gap Δ. Fast *non-mixing* hidden dynamics — conserved circulation — retain records with no timescale separation at all (`fastbath_probes.py`, [Main §1.3]), which is why (C2) is stated in two forms and this necessity belongs to the slow form.

Necessity of C3 (sufficient capacity). The non-Markovian mutual information is bounded by the hidden sector's size:

$$I(X_{<t}; X_{>t} \mid X_t) \leq \log_2 m$$

where m = |C_H|. Proof: the total system is deterministic, so X_{>t} is a function of (X_t, H_t). Given X_t, the chain X_{<t} → H_t → X_{>t} is Markov. By the data processing inequality:

$$I(X_{<t}; X_{>t} \mid X_t) \leq I(X_{<t}; H_t \mid X_t) \leq H(H_t \mid X_t) \leq H(H_t) \leq \log_2 m$$

Each step uses a standard information-theoretic inequality. The result: if you want K bits of history dependence, you need m ≥ 2^K hidden states.

The complete characterization. For |C_V| ≥ 2: the process is non-Markovian on accessible timescales ⟺ for some accessible order k it arises, on each finite accessible horizon K ≥ k+1, from marginalizing a deterministic bijection under C1, C3, C4, with the readback at k (C2 sits outside the equivalence: the existential realization satisfies it trivially with a static bath). Both directions are the foundational paper's own theorems; every finite-horizon law admits the fixed-basis unitary quantum representation ($S \iff D \iff Q_{\text{fb}}$, internal — the compressed form for the correspondence's class under (T)), and the fixed-$\hat{H}$ form is constructive for every realized process. No two of representability, memory and P-indivisibility are equivalent: representability alone does not force memory (diagonal Hamiltonians), and conditional memory alone does not force P-indivisibility (the XOR family).

What "unitarily evolving QM" means precisely. At the finite observable-law layer the framework gives an exact fixed-basis Hilbert/unitary/Born representation: $S \iff D \iff Q_{\mathrm{fb}}$. This is not metaphorical, and it includes exact multi-time fixed-basis records through the permutation-unitary realization. It does **not** by itself derive the whole operational category of quantum mechanics. The visible–hidden tensor factor used by the dilation is part of the representation construction; for two spatial laboratories, the OI dependency graph supplies exact causal cones, but a common local tensor/composite instrument structure is part of the coherent operational lift, characterized in Main §3.4 and GR §3.3 by five operational conditions. Conditioning on a registered classical outcome is represented by ordinary conditioning on the substratum ensemble; identifying every coherent quantum update with that construction is likewise part of the operational lift. The memory-bearing sector ($M_t > 0$) carries the specifically OI content.

And it does not fix which unitary relates two times. The equivalence says which visible laws have a fixed-basis unitary representation. It does not pick out the evolution carrying one time to the next. For a dilated process, one visible family can admit several coherent unitary lifts — each reproducing the visible law at every time, each composing exactly across times — whose relating evolutions differ, because a lift can be deformed by a time-dependent right action that preserves everything the visible law sees. A deformation that is constant in time cancels from every relating evolution, so the nonuniqueness is specifically a time-dependence phenomenon. Uniqueness therefore needs a selection condition beyond visible-law representability and ordinary coherence, and none is claimed or named here. The invisible freedom itself has two sides — a right action by the anchored class and a left action within each visible output's hidden fibre, each maximal among uniform actions of its kind — and it does not exhaust the freedom: two lifts can fail to be related by any uniformly invisible right deformation and still agree on every relating evolution, and a pair outside that right class can differ in relating evolution while being related by a left in-fibre move. At one time, what the visible law leaves free after both actions is known exactly: it is the off-diagonal part of the fibre-Gram data — for each visible output, the overlaps among the hidden states into which the lift sends the visible inputs — modulo phases, with the visible law as its diagonal; and that residue is not inert, since two lifts of one visible law can differ in it and differ in relating evolution. That residue does not reach the fixed-basis layer. With a trivial ancilla, the fixed-basis datum built from a lift's unitary at one time, with uniform initial law and identity readout, has finite-horizon trajectory laws fixed by the visible law alone, also after contractively scaled partial permutations on either side, and two such data have equal laws exactly when their visible laws are equal (kernel: `bridge_traj`, `rooted_eq_iff_slice_eq`); the fixed-basis correspondence acts on the quotient by visible equality. The trivial ancilla is part of that statement: an ancilla attached as a tensor factor with its own unitary leaves the rooted family unchanged, while an ancilla carried between steps as hidden basis states can separate two lifts that share every column

entering the visible law, already in the two-step law. What remains open is two-part: what selects the trajectory of that data through time, and what fixes the relating evolution within a trajectory — the latter not fixed even by the complete trajectory. No selection principle is named or excluded. This leaves $S \iff D \iff Q_{\mathrm{fb}}$ exactly as stated and does not make embedded observation and quantum mechanics inequivalent.

The bridge therefore has three statuses, and they are worth stating together. Finite observable-law representability is **settled**: every finite-horizon law has the fixed-basis unitary/Born representation, and the three descriptions are equivalent. Exact finite operational quantum mechanics is **characterized conditionally**: it holds exactly when the five completion conditions hold, none of them supplied by embedded observation alone. And selecting one relative quantum evolution from the coherent lifts compatible with a single visible family is **open** — the remaining bare-OI question, on which ordinary coherence has been shown insufficient and nothing further is claimed.

A note on what is proved versus imported, to keep the claim honest. The exact finite-law representation is internal: every finite stochastic law has a reversible deterministic realization and an exact fixed-basis unitary/Born representation ($S \iff D \iff Q_{\mathrm{fb}}$, [Main §3.4]). That statement is universal — ordinary Markov chains are included — so representability itself is not the discriminator. C4 supplies the nontrivial content: accessible history readback and unavoidable hidden predictive memory. The Barandes stochastic–quantum correspondence remains useful for a compressed representation and its interpretive apparatus; it is not needed to prove that some Hilbert/Born representation exists. The remaining foundational qualification is narrower than a selection claim: the coherent preparations, noncommuting instruments, and composites of one OI system form one standard quantum operational theory exactly when five operational conditions hold — valid probabilities, trivial-ancilla consistency, inert spectators, full reversible control, and iterated composition — none supplied by embedded observation alone, so the framework classifies the quantum completions rather than deriving quantum mechanics outright (Main §3.4; GR §3.3). The layered form of that classification, quantum-complete OI (OI^{+}) — the unchanged OI core with well-formedness, observational independence, reversible richness, and observer recursion — is equivalent on every nonempty finite carrier to exact finite endomorphic operational quantum mechanics, with no claim that bare OI entails the added principles (GR §3.3). A primitive-source refinement restates those principles at the level of physical implementations and embedded observers — implementation locality, phase-free richness — one continuously driven transition and the exchanges of distinguishable states, with no phase operation — and embedded observation, with no closure of the implementations under the adjoint and no inverse accessibility assumed — equivalent to the same operational quantum mechanics on every nonempty finite carrier (GR §3.3). A substratum-source form completes that refinement: the concrete OI substratum supplies the structural architecture — closure under composition and coarse-graining, stability under an uncoupled spectator, relabelling invari-

ance, reversibility, the exchanges of distinguishable states, and, as a stated intervention principle rather than a consequence of its finite states and bijective dynamics, the phases — but, under finite bijective read-write dynamics, not the continuously tunable state-mixing operation; with that one operation added as an empirical controllability resource it is equivalent to the same operational quantum mechanics on every nonempty finite carrier, while finite reversible read-write dynamics, even with genuine hidden-memory and readback behavior, does not itself generate quantum state mixing (GR §3.3). Its typed form removes the endomorphic qualifier: under the carrier-general typed operational interface — identity, composition, coarse-graining, relabelling, uniform finite attachment, discard, and factor readout, an interface that itself carries no quantum content — the same equivalence holds with exact finite-dimensional typed operational quantum mechanics, operations between any two nonempty finite carriers included, so the remaining qualifier is finite-dimensional rather than endomorphic (GR §3.3). The finite theory admits a canonical infinite-region completion without a continuum limit: the finite regions of the fixed-spacing lattice carry full matrix algebras joined by compatible injective unital embeddings with observables of disjoint regions commuting, the norm completion of their union is a quasilocal C^*-algebra on which every consistent family of finite-region density matrices is a state and the reversible finite-range substratum update is an isometric $*$-automorphism, and any system carrying the same local stages is canonically and uniquely $*$-isomorphic to it, intertwining the dynamics; this removes the finite-region cutoff for the algebra, the state space, and the OI-induced discrete dynamics rather than characterizing all infinite-dimensional quantum instruments or all locality-preserving quantum dynamics; the lattice spacing stays fundamental, no Hilbert-space representation and no superselection sector is selected at the level of the laws, the restriction to the substratum-induced dynamics is a theorem rather than a convention — a phase automorphism at one site preserves locality without arising from any reversible finite-range bijection — and, while the update is joined to the identity by an exact norm-continuous path of local $*$-automorphisms of that algebra, with no one-parameter-group law established and no generator exhibited, a continuous-time Hamiltonian law remains additional structure (GR §3.3). Within the structural architecture the substratum supplies, with the phase intervention stated, exact finite endomorphic operational quantum mechanics holds exactly when one continuous layer flow of an involution of the configuration space moving some configuration is an available operation at every level and every intermediate time, the executability of that flow being an assumption not derived by configuration-level generation, as the phase intervention is an assumption on the substratum class and not a consequence of the finite states with the bijective dynamics (GR §3.3). The stated substratum dynamics with the observer's read and write access do not source that executability, the layers being available at time one and not at intermediate times, and the observer-level lift, in every formulation stated that lands in the operational interface, derives neither a non-monomial admissible operator nor the layer flow, the site-space wave-operator formulation being unresolved rather than excluded (GR §3.3). Bell is kept separate from that operational-lift residue:

the response-complete deterministic layer retains measurement independence and requires ontic parameter dependence for quantum Bell violation, while the stochastic causal-local Tsirelson result remains a weaker-description statement. Thus the accurate headline is strong but layered: the finite observable stochastic/reversible/quantum descriptions are exactly equivalent; readback identifies the memory-bearing OI sector; the common coherent local operational lift is stated separately rather than silently folded into the equivalence theorem.

Passive observation, kept apart. Four finite-dimensional statements about passive observation stand beside that classification and enter none of its conditions (Main §3.4). Complete passive observation of an observable algebra in block-diagonal form is possible exactly when the algebra is commutative — some passive instrument separates its states exactly then — and every passive instrument induces a classical stochastic observation of the center: quantum non-commutativity cannot coexist with complete passive observation. The fact is theory-insensitive — passive incompleteness holds in every finite operational theory on two or more states, so it carries no information about whether the OI core is realized; the implication from the core holds vacuously and its converse fails — and it is therefore not evidence for a hidden ontology, nor a statement that quantum mechanics rests on observation incompleteness. A passive internal observer can only read an existing record, and creating a genuinely new record changes the joint system.

The uncertainty principle. The uncertainty relations of quantum theory — Robertson's preparation inequality $\sigma_A\sigma_B \geq \frac{1}{2}|\langle[A,B]\rangle|$ and the error–disturbance relations — are inequalities in the commutator, and the passive-observation statement is a qualitative finite-dimensional consequence of the same non-commutativity: an instrument that leaves every state unchanged on average learns only about the center of the observable algebra. Like passive incompleteness, the preparation inequality holds in every finite operational theory, with or without the OI core. Observational incompleteness in the sense of the OI core is a different statement, about the hidden sector of an embedded observer: under C4 it is present already for a single fixed readout, where no choice between measurements arises, and the visible laws of fixed-readout processes are exactly the finite stochastic laws, which no uncertainty relation constrains. The relations bind only where non-commuting observables are operationally available, and that availability belongs to the completion question: the non-commutative operational structure follows from the completion principles of OI^+, with the OI-core clause redundant once reversible control is present, and whether embedded observation selects it is open (Main §3.4).

The Cosmological Application (§4)

4.1 The Partition

The cosmological horizon is the boundary beyond which no signal traveling at or below c can ever reach the observer. In a universe with a positive cosmological constant (de Sitter space), this horizon exists for every observer and has a definite, finite area:

$$A = \frac{4\pi c^2}{H^2}$$

where H is the Hubble parameter. This implements Lemma 2 naturally: Γ_V = everything inside the horizon, Γ_H = everything beyond.

Why $\hbar$ is the same for all observers: Different observers have slightly different horizons. But the gap equation $\hbar = c^3\varepsilon^2/(4G\sigma)$ depends only on local quantities (c, G, ε, and the entropy σ of one boundary mode) — not on the horizon area or the observer's worldline. So all observers derive the same $\hbar$.

4.2 Verification of the Conditions

C1 (coupling). In general relativity's ADM formulation, the bulk Hamiltonian is a sum of constraints that vanish on-shell — meaning the "real" dynamics happens at the boundary. The Hamiltonian and momentum constraints correlate interior and exterior data. This is stronger than just H_int $\neq 0$: the constraints enforce correlations that persist on all timescales.

C2 (memory persistence). τ_B $\sim 1/H \sim 10^{17}$ seconds (the Hubble time). τ_S $\sim 10^{-15}$ seconds (a typical atomic process). The ratio: τ_S/τ_B $\sim 10^{-32}$.

C3 (sufficient capacity). The hidden sector has $A/\varepsilon^2 = S_dS/\sigma \sim 10^{122}$ modes at $\sigma = 1$ (S_dS the de Sitter entropy). The visible sector has $\sim 10^{80}$ baryons. No experiment comes close to saturating the hidden sector.

4.3 Application

With the cosmological horizon satisfying the definition and all conditions, the characterization theorem applies. The observer's reduced description is non-Markovian — P-indivisible in the recurrence regimes — and admits a unitarily evolving quantum representation ([Main §3.4]). The action scale $\hbar$ is fixed by the partition geometry in terms of the discreteness scale — which is what Section 5 derives.

Where Planck's Constant Comes From (§5)

Partition-relativity proved that, under the canonical measure, the emergent quantum description is completely and uniquely determined by the partition.

This means $\hbar$ cannot be a free parameter — it must be fixed by the geometry of the boundary.

The Classical Horizon Temperature (§5.1)

Starting point: Jacobson's identity. Jacobson (1995) showed that Einstein's field equations follow from applying the first law of thermodynamics δQ = TdS to local causal horizons:

$$dE = \frac{c^2\kappa}{8\pi G}\, dA$$

where κ is the surface gravity and dA is the change in horizon area. This is a classical gravitational identity — no quantum mechanics involved.

The entropy density is $\eta = \sigma/\varepsilon^2$ — one coupled mode per minimal cell of area ε^2, each carrying an entropy σ. ε is defined as the minimal distinguishable scale (Lemma 1), so each cell of area ε^2 contributes exactly one boundary mode that couples across the partition. The entropy σ of a mode is a pure number, in nats: the number of internal states per mode (the alphabet size q) bounds it, $\sigma \leq \ln$ q, and the alphabet-size hypothesis [SM §2.7] does not fix it. The Bekenstein-Hawking coefficient does not depend on σ, which cancels from the entropy law and enters only the discreteness scale, $\varepsilon = 2\sqrt{\sigma}$ l_p [GR §3.1, §4]. So dS = σ dA/ε^2. From dE = TdS:

$$k_B T_{\mathrm{cl}} = \frac{c^2\epsilon^2\kappa}{8\pi G\sigma}$$

For the de Sitter horizon, κ = cH:

$$k_B T_{\mathrm{cl}} = \frac{c^3\epsilon^2 H}{8\pi G\sigma}$$

Critical point: no $\hbar$ appears anywhere. This temperature is computed from purely classical quantities.

The Four-Step Derivation (§5.2)

Step 1 (Uniqueness). Partition-relativity guarantees $\hbar$ is determined by the partition geometry. It's not a free parameter.

Step 2 (Boundary-only dependence). A substantial lemma showing that $\hbar$ depends only on the boundary modes, not on the deep interior of the hidden sector. Decompose the hidden sector into boundary modes C_B (near the horizon) and deep modes C_D (far from the horizon):

$$\mathcal{C}_H = \mathcal{C}_B \times \mathcal{C}_D$$

The proof has three parts: (i) spatial locality means V talks to B, and B talks to D, but V doesn't talk directly to D; (ii) on timescales t $\ll \tau$_B, the deep sector barely moves (displacement $\sim 10^{-32}$); (iii) because the deep modes are frozen, the sum over them produces a trivial factor:

$$T_{ij}(t) = \underbrace{\frac{1}{|\mathcal{C}_B|}\sum_b \delta_{x_j}[\pi_V(\varphi_t^{VB}(x_i,b))]}_{T_{ij}^{(B)}(t)} + \mathcal{O}(t/\tau_B)$$

The transition probabilities depend only on boundary dynamics. Since $\hbar$ is determined by transition probabilities, $\hbar$ depends only on boundary quantities: c, G, and ε. This excludes dependence on H — if $\hbar$ depended on H, observers at different cosmic epochs would have different quantum mechanics.

Step 3 (Dimensional analysis). Step 2 excludes volumetric (deep-sector) quantities, leaving boundary quantities. The boundary carries both *local* geometric data (ε, κ, and the constants c, G) and a *global* quantity: the total area A, which forms the dimensionless ratio A/ε^2 = S_dS/σ. If $\hbar$ depended on S_dS, it would be observer-dependent — different observers have different horizon areas — contradicting the universality of the emergent action scale. This excludes A. The surface gravity κ is excluded by two independent arguments: (i) κ differs between horizon types (cosmological vs. black hole), but $\hbar$ is universal — a laboratory experiment measures the same $\hbar$ regardless of which horizon defines the partition; (ii) for the cosmological horizon, κ = cH is time-dependent ($\dot{H} \neq 0$), but $\hbar$ is observed to be constant on laboratory timescales. The unique combination of c, G, and ε with dimensions of action:

$$[\hbar] = \frac{[c]^3\,[\epsilon]^2}{[G]} = \mathrm{kg}\cdot\mathrm{m}^2/\mathrm{s} \quad \checkmark$$

So $\hbar = \beta\ \mathrm{c}^3\varepsilon^2/\mathrm{G}$, where β is a dimensionless constant that dimensional analysis alone can't fix.

Step 4 (Thermal self-consistency). We have two independent descriptions of the horizon temperature:

Classical: T_cl = $\mathrm{c}^2\varepsilon^2\kappa/(8\pi\mathrm{Gk_B}\sigma)$ — computed from the geometric substratum, no $\hbar$.

Quantum: The emergent QFT lives on a spacetime with a bifurcate Killing horizon. Standard QFT on curved spacetime gives a KMS thermal state at temperature:

$$T_Q = \frac{\hbar\kappa}{2\pi c k_B}$$

The two temperatures are computed independently — $T_{\rm cl}$ from the classical substratum alone (no QM), T_Q from the emergent QFT alone (no classical substratum details) — but they describe the same physical degrees of freedom: the boundary modes across which the partition is defined. Since the quantum description is derived from the classical one (Part I), and the derivation is exact at the boundary, the two descriptions cannot assign contradictory temperatures. Consistency requires $T_{\rm cl} = T_Q$:

$$\frac{c^2\epsilon^2\kappa}{8\pi G\sigma} = \frac{\hbar\kappa}{2\pi c}$$

The surface gravity κ cancels from both sides. Solving:

$$\boxed{\hbar = \frac{c^3\epsilon^2}{4G\sigma}}$$

This fixes $\beta = 1/(4\sigma)$.

Why this is not circular. The KMS temperature T_Q contains $\hbar$ as an *unknown*. The classical temperature T_cl contains no $\hbar$ at all. The non-circularity is structural: Part I establishes that a QFT emerges with *some* action scale $\hbar$; §5 determines *which* $\hbar$, using the independent classical temperature that Part I neither requires nor produces. If T_cl had depended on the deep hidden-sector volume (it doesn't — the boundary-only lemma excludes it), or if T_Q had been state-dependent (it isn't — the KMS temperature is kinematic in its parameter, conditional on H-balance in its class), the matching wouldn't work. That neither pathology obtains makes the gap equation a genuine determination.

What the gap equation fixes. The gap equation, conditional on the infrared detailed-balance condition H-slope with the horizon and frame conditions [GR §3.2], relates one free geometric parameter (ε), at given σ, to one output ($\hbar$). The specific relationship $\hbar = c^3\varepsilon^2/(4G\sigma)$ enters three later results — the Bekenstein-Hawking factor 1/4 (§6), the compression ratio of the conditional CC dissolution (§7.3), and the Type II RVM dark-energy reading (§8.1). The three share the calibration; they are consequences of it, not independent checks of it. See Appendix item 1 for the detailed discussion.

Frequencies and the D-Gauge Question (§5.3)

The problem. The relation $T_{ij} = |U_{ij}|^2$ discards phase information. Could different Hamiltonians give the same transition probabilities?

What is proved (Bohr-frequency completeness). The transition probabilities determine the full set of energy differences $E_k - E_l$. Each return probability expands as $|U_{ii}(t)|^2 = \Sigma\ |V_{ik}|^2|V_{il}|^2\ e^{-i(E_k-E_l)t}$, in

which every gap carries a strictly positive coefficient — degenerate gaps merge coefficients but cannot cancel them — and distinct frequencies give linearly independent characters, so equal probability families have equal gap sets.

Conditional two-branch reconstruction. In dimension at least three, under G1–G3 for both Hamiltonians and the global sign-aware integer classification H-BG of [GR §3.3], matching probabilities imply H' = DHD† + E_0 or H' = −DHD† + E_0. H-BG allows the exceptional six-mark correspondence or its global reversal; its complete comparison with the published classification remains open. The unsigned formal premise `BGIntegerClassification` is refuted by reflecting one of the homometric six-mark rulers. Each displayed ambiguity is exactly invisible. The C_4 ring of [SM] Proposition 20 has degenerate gaps; its conjugate H reproduces every transition probability and lies in the reflection branch, H = −D_s H D_s† with D_s = diag(1, −1, 1, −1), so this exhibited partner does not show a failure, without the gap condition, of the two-branch form conditional on H-BG. Bohr-frequency completeness, the real-to-integer realization transfer, congruent reconstruction and the positive six-mode obstruction are established independently; they do not by themselves prove the complete classification. The double-difference condition

$$\delta_{ik} - \delta_{il} - \delta_{jk} + \delta_{jl} = 0 \pmod{2\pi}$$

with general solution δ_ik = α_i + β_k then closes the rephasing factor once the mode correspondence is pinned.

The dimensional obstruction. The unitary U(t) is dimensionless. The Hamiltonian $\hat{H} = i\hbar\,\partial_t U \cdot U^\dagger$ contains ħ, which is dimensionful. No amount of dimensionless data can fix a dimensionful constant. This is why Step 4 (thermal matching) is not just a convenient check but the *mathematically obligatory* step: it's the only place where dimensionful physical input enters the framework.

The Discreteness Scale (§6)

What $\varepsilon = 2\sqrt{\sigma}$ l_p means

Rearranging the gap equation $\hbar = c^3\varepsilon^2/(4G\sigma)$:

$$\epsilon^2 = \frac{4\sigma\hbar G}{c^3} = 4\sigma\, l_p^2$$

where $l_p = \sqrt{(\hbar G/c^3)}$ is the Planck length. Therefore $\varepsilon = 2\sqrt{\sigma}$ l_p, which is ε = 2 l_p at σ = 1 (one nat per boundary mode).

What this is: The framework has one free geometric parameter — the discreteness scale ε. The gap equation fixes its relationship to ħ. Given the measured

value of $\hbar$ and the entropy σ per mode, the discreteness scale is determined: $\varepsilon = 2\sqrt{\sigma}$ l_p, which at $\sigma = 1$ is 2 l_p $\approx 3.2 \times 10^{-35}$ meters.

What this is NOT: This is not an independent prediction of ε. The framework contains one free geometric parameter and one equation relating it to known constants and σ. The self-consistency condition fixes the ratio $\varepsilon/\sqrt{\sigma} = 2$ l_p, which puts ε in the Planck regime only if σ is of order one, and doesn't predict a number that wasn't already implicitly known.

The Bekenstein-Hawking Entropy — Where the 1/4 factor comes from

The number of independent modes on the cosmological horizon is:

$$N_{\text{modes}} = \frac{A}{\epsilon^2} = \frac{A}{4\sigma\, l_p^2}$$

Each mode carries the entropy σ, so the total entropy is the Bekenstein-Hawking entropy:

$$S_{\text{BH}} = \sigma N_{\text{modes}} = \frac{A}{4\, l_p^2}$$

The factor of 1/4 — which Bekenstein introduced in 1973 with a coefficient he could estimate only heuristically, and which Hawking fixed in 1975 from the Hawking temperature and the first law — is here recovered from the cell size: each minimal cell of area $\varepsilon^2 = 4\sigma$ l_p^2 contributes the entropy σ of one boundary mode, σ cancels, and the 4 in the denominator comes from $\varepsilon^2 = 4\sigma$ l_p^2, which the gap equation supplies under one named condition on the horizon reservoir (H-slope, the infrared detailed-balance condition of [GR §3.2]) together with the horizon and frame conditions of [GR §2, §8.5].

What this shows: The standard derivation fixes the 1/4 from the Hawking temperature and the first law. The framework reproduces the same ratio, $2\pi/8\pi$, from its horizon calibration with no fitted input, under the named horizon conditions — a consistency relation with Hawking's result rather than an independent test of it; the counting itself, S = σA/ε^2, needs none of the conditions.

What self-consistency does not bound

The gap equation fixes $\varepsilon/\sqrt{\sigma} = 2$ l_p, not ε by itself. ε sits at the Planck scale only if σ is of order one. The count caps σ at ln q for a mode with q states but does not bound it from below: $\sigma = 10^{-12}$ would give $\varepsilon = 2 \times 10^{-6}$ l_p. The gap equation does not exclude scales far from l_p; whether they are excluded on independent grounds is open.

The Two Levels

Now the framework's conditional treatment of the cosmological constant problem. But first, a crucial link: how does finite-dimensional quantum mechanics become quantum *field* theory — with independent modes, each carrying zero-point energy?

The answer is spatial locality. The classical substratum has it — neighboring cells interact, distant cells don't. The paper proves that the emergent quantum description inherits it. The argument is direct: the Barandes correspondence maps each classical configuration $(x_1, x_2, \ldots, x_N)$ to a quantum basis state $|x_1, x_2, \ldots, x_N\rangle$. That's already a tensor product. The only question is whether the emergent Hamiltonian respects the structure. And it must, because if two configurations differ only at sites that aren't neighbors, the classical dynamics can't connect them in an infinitesimal time step — so the transition probability between them is zero — so the corresponding quantum Hamiltonian matrix element is zero. The emergent Hamiltonian couples only neighboring cells, exactly like the classical one. You get a lattice quantum field theory with a built-in ultraviolet cutoff at $\epsilon = 2\sqrt{\sigma}\, l_p$.

This is what makes the CC dissolution work. The QFT has independent modes. Each mode gets a zero-point energy. The sum diverges. And the "worst prediction in physics" follows — but only within the emergent description.

The critical insight is that general relativity and quantum field theory are not competing answers to the same question. They are answers to *different questions*, asked at different levels of the same reality.

Level 1: The classical substratum. Spacetime geometry is part of the fundamental layer. The metric tensor evolves via Einstein's field equations. The stress-energy tensor that sources gravity is the classical stress-energy of the total microstate. At this level, the vacuum energy density sits at the critical scale: $\rho \sim H^2/G \sim 10^{-9}$ J/m^3. No zero-point energy. No discrepancy.

Level 2: The emergent quantum description. For an embedded observer tracing out the hidden sector, the mandatory quantum description assigns a zero-point energy of $\frac{1}{2}\hbar\omega$ per mode. Sum to the Planck cutoff and you get $\rho_{\text{QFT}} \sim 10^{113}$ J/m^3. This number is real *within the quantum description* — it reflects the magnitude of the trace-out noise — but it is not a source term in Einstein's equations, because those equations operate at the classical level, which is logically prior to the quantum description.

This ordering is not a choice but is forced by three independent requirements. First, the partition must be definite — not in superposition — for the trace-out to be well-defined; a partition in superposition would yield an incoherent mixture of inequivalent quantum theories. Second, the partition is defined by the causal structure, which is determined by null geodesics of the metric; if the metric were derived from QM, the derivation would be circular (QM $\to$ metric $\to$ causal structure $\to$ partition $\to$ QM). Third, $\hbar$ is determined by the boundary

geometry; if the geometry were itself quantum-mechanical, $\hbar$ would depend on a quantum state, contradicting its observed universality.

On the framework's reading, the 10^{122} ratio between the two answers is not a discrepancy. Up to a convention-dependent factor it equals $S_{\rm dS}$ — the Bekenstein-Hawking entropy of the cosmological horizon — as it does for any theory with a Planck cutoff and a Hubble-scale density; the framework reads it as the entropy of the hidden-sector degrees of freedom the trace-out compresses into the emergent quantum state. On that reading, conditional on the open coupling step, the "worst prediction in physics" is the information compression ratio of the observer's blind spot: a category error, not a fine-tuning failure.

The observed vacuum energy, measured since 1998, is of the Friedmann scale H^2/G, as any $O(1)$ component of a flat universe is; its value is set by initial conditions, so this is a normalization, not a prediction. Meanwhile, every attempt to search for a cancellation mechanism within quantum-first frameworks has found nothing. If the open coupling step holds, the framework explains why: there is nothing to cancel.

This track record is itself evidence for the ordering. The question of whether geometry is prior to quantum mechanics or quantum mechanics is prior to geometry is usually treated as a philosophical preference. But it has an empirical signature sitting in plain sight: one ordering produces the worst prediction in physics and has no solution; the other predicts the observed value and explains the discrepancy as a derived quantity. That doesn't prove the ordering is correct, but it's existing evidence, not a future hope. (Note the division of labor with the non-adjudicability result in the gravity paper: two descriptions that fit the same observations equally well can never be ranked from inside — that is the factoring argument — but the two *programs* here are not in that situation. One predicts the observed vacuum energy and one misses it by 10^{122}; predictive asymmetry on accessible data is exactly the kind of criterion the factoring argument leaves open.)

What Quantum Weirdness Looks Like From Here

Feynman opens his quantum mechanics volume with a famous warning. The double-slit experiment, he writes in Volume III, Lecture 1 of the *Lectures on Physics*, is "a phenomenon which is impossible, absolutely impossible, to explain in any classical way, and which has in it the heart of quantum mechanics. In reality, it contains the only mystery." He goes on: "We cannot make the mystery go away by 'explaining' how it works. We will just tell you how it works."

If quantum mechanics is an emergent description forced on embedded observers, the standard quantum puzzles acquire straightforward readings — and the "only mystery" stops being mysterious. It becomes a structural consequence.

The double-slit experiment. The particle goes through one slit. The interference pattern arises because the transition probabilities are computed by marginalizing over the hidden sector, and the hidden sector includes the field configuration near both slits. Opening or closing the second slit changes the boundary conditions, which changes the marginalization, which changes the pattern. The "wave-like" behavior is the hidden sector's influence shifting when the geometry changes. The mystery dissolves not into another postulate but into a derivation: this is what marginal statistics look like when paths through the partition are indistinguishable.

Entanglement. Two particles prepared together share a joint transition structure inherited from the trace-out, but Bell violation cannot be explained merely by calling that reduced law indivisible. At the response-complete deterministic level, measurement independence plus parameter independence would force Bell factorization and CHSH ≤ 2. The framework therefore gives up **ontic parameter independence**: the microstate response of one wing may depend on the remote setting, while the averaged operational statistics remain no-signaling. Superdeterminism — ontic measurement dependence — is the alternative it records and declines. The stochastic indivisible construction's Tsirelson theorem remains a separate reduced-law result.

The measurement problem. Measurement produces definite outcomes through the indivisible dynamics. When Wigner can't access his Friend's lab, he traces out its internal degrees of freedom and assigns a superposition. The superposition reflects Wigner's epistemic situation — what *he* can infer — not the Friend's physical state. Branching in the Many-Worlds sense is a feature of the compressed description, not of the underlying reality.

Schrödinger's cat. The cat is always alive or always dead — the substratum holds exactly one configuration at every moment, and nothing in the box is smeared. The superposition is the *outside observer's* ledger: the record of which outcome occurred has been written into degrees of freedom the observer cannot read, so the honest description from outside prices both branches, the way the coin-and-die model (§2.4) prices its unobserved intermediate step. And the famous absurdity dissolves on a second look: a real cat is never an unobserved step. Air molecules, thermal photons, the Geiger counter itself — the environment reads the record continuously, and each interaction forces the statistics back to ordinary either/or probabilities. That is why no consciousness was ever needed to open the box; one stray photon is observer enough. Schrödinger built the scenario as a reductio — he thought a theory implying alive-and-dead cats had gone wrong somewhere — and the framework agrees, while relocating the error: not in the dynamics, which needed no collapse law added, but in reading the observer's compression algorithm as the state of the cat.

The common thread across the three readings is that what looked like irreducible quantum mystery is the partition's reflection of substratum dynamics. You're embedded in the universe; you don't get the view from nowhere; the description you arrive at is constrained by what marginal access permits. Feynman taught

his students that the rules of QM "had to just be accepted." The framework's claim is that the rules don't have to be accepted — they have to be derived, and the derivation makes contact with the fact of embedding. Whether the derivation is correct is a separate question; the empirical predictions that distinguish it from competing accounts are listed below.

Predictions

The framework is not just a reinterpretation. It makes specific, falsifiable predictions — and, crucially, it makes predictions in domains it was not designed to address.

Dark energy evolution. Because the hidden sector's dimensionality changes as the Hubble parameter H evolves (the horizon area is $A = 4\pi c^2/H^2$), the emergent vacuum energy inherits a dependence on H. The resulting form matches the Running Vacuum Model: $\Lambda_{\text{eff}} = \Lambda_0 + \nu H^2$. The framework's *structural* commitments — boundary entropy depends on H, matter is conserved in the trace-out, and Bianchi consistency then forces $G(H)$ to run in tandem with $\rho_\Lambda(H)$ — select the Type II channel, a structural reading rather than a derived law; the functional form itself is the local expansion shared by every smooth dark-energy model ([GR §7.1]), so the content is the reading and the magnitude. The magnitude of ν is bounded by the emergent-QFT calculation (conditional on the [GR §7.3] split, not established at theorem level): applying standard curved-spacetime running to the Standard Model fields at their physical masses on the derived classical background yields $|\nu_{\text{QFT}}| \sim (M_{\text{SM}}/M_{\text{Pl}})^2 \sim 10^{-32}$ — observationally indistinguishable from zero. A potential additional contribution from the substratum mode count is constrained by the boundary architecture: modes are catalogued by spatial location alone (one mode per cell of area ϵ^2, no inherent frequency assignment), so a natural 2D distribution concentrates them near the UV cutoff and contributes well below $|\nu_{\text{QFT}}|$. Absent additional substratum structure giving a log-uniform boundary-mode distribution at the cosmological horizon, the framework's expectation reduces to $|\nu| \approx |\nu_{\text{QFT}}|$. A direct test is provided by Bertini, Novaes, von Marttens, and Shapiro (arXiv:2509.24026), who fit a renormalization-group-corrected ΛCDM model — structurally equivalent to Type II RVM at leading order, with both $G(\mu)$ and $\Lambda(\mu)$ running and matter conserved — against Planck 2018 + DESI DR2 + DES-Y5, reporting $\nu = -(2.5 \pm 1.3) \times 10^{-4}$, consistent with zero at 2σ and consistent with that expectation. The structural content — the smooth geometry-dependent bookkeeping read in the Type II channel — is a framework commitment at the level of a reading, not a derived law. DESI Year 5 ($\sim 1\%$ precision on ν) is the decisive test of that reading: a null at that precision supports it; a high-significance detection of $|\nu|$ at the 10^{-4} level would require additional substratum structure beyond what the present boundary-entropy construction supplies — structure a direct computation of the boundary spectral measure from the substratum dynamics now shows the linear map does not generate (the measure is power-law

local, never log-uniform), so such a detection would also demand its correlated Lorentz-violation co-signature.

The dark sector as corroboration. The trace-out that produces the memory-bearing dynamics quantum mechanics represents has a gravitational consequence that the paper did not set out to find. The boundary entropy — the entropy $S_{\rm dS}$ of the A/ϵ^2 modes traced out to produce the quantum description — has thermal energy that, distributed over the Hubble volume, equals the critical density exactly. A crucial subtlety: this thermal energy is computed entirely from pre-trace-out (classical) quantities — the boundary entropy, the classical horizon temperature, and the Hubble volume — with no reference to $\hbar$ or the emergent quantum description. This is what distinguishes the boundary entropy's gravitational contribution from the QFT zero-point energy ($\rho_{\rm QFT} \sim 10^{113}$ J/m^3), which exists only *after* the trace-out and is an artifact of the emergent description. The framework denies that the zero-point energy gravitates (this is the CC dissolution of Part II); the boundary entropy's classical thermal energy *does* gravitate, at the scale $\rho_{\rm crit} \sim 10^{-9}$ J/m^3.

The paper proves that this entropy has no operator in the emergent QFT. The baryonic sector — what QFT can account for — is ~5% of $\rho_{\rm crit}$. The remaining ~95% is the boundary entropy: gravitationally active, invisible to the emergent description, and persistent (condition C2). This matches the observed composition of the universe, in which ~95% of the gravitational content has no source in particle physics. The uniform component corresponds to dark energy (handled by Part II's CC dissolution). The structured component — dark matter — arises from matter-induced entropy displacement: baryonic matter displaces boundary entropy via the Clausius relation, the Jacobson mechanism converts the entropy gradient into curvature, yielding the MOND acceleration scale $a_0 = cH/6 \approx 1.1 \times 10^{-10}$ m/s^2 and the baryonic Tully-Fisher relation $v^4 = GM_B \cdot cH/6$ — both parameter-free once the dimensional factor 1/6 is imported from Verlinde's analysis (GR, §7.3). That a definition designed to formalize observation also account for the dark sector's total budget and internal structure is independent corroboration that observational incompleteness is capturing real structure.

High-redshift dark matter. Because $a_0(z) = cH(z)/6$ and $H(z)$ increases with redshift, the dark matter phenomenology evolves: a_0 is 1.8× larger at $z = 1$, 3.0× at $z = 2$, and 4.6× at $z = 3$. This shrinks the MOND crossover radius, so at a fixed baryonic mass profile a galaxy at high redshift is less baryon-dominated and its rotation curve reaches the flat deep-MOND form at smaller radii. Two observations run against this. Genzel et al. (Nature, 2017) report stacked rotation curves at $z = 0.9$–2.4 with outer velocities declining at $> 3\sigma$ significance relative to local spirals. In the Milky Way, Jiao et al. (*A&A* 678, A208, 2023) detect a Keplerian decline from ~ 19 to ~ 26.5 kpc using Gaia DR3 kinematics, with the flat rotation curve hypothesis rejected at 3σ; that range lies beyond the crossover ($r_M \approx 9$–16 kpc, depending on the baryon budget), where the law gives a nearly flat curve. Both are in tension with

the law; a quantitative comparison needs the baryon models, at high redshift the galaxies' own baryon profiles and pressure support, and the observers' own velocity analyses ([GR §7.3]). The baryonic Tully-Fisher relation also evolves: $v_{\rm flat} \propto H(z)^{1/4}$, predicting 32% higher velocities at $z = 2$ at fixed baryonic mass. McGaugh et al. (2024) find no clear evolution in the *baryonic* TF to $z \sim 2.5$, with unresolved differences between samples in baryonic-mass definitions and velocity measurements. The law predicts a baryonic-mass offset of -0.48 dex at fixed velocity at $z = 2$, so this result too is a tension with $v^4 \propto H(z)$, not yet a statistically quantified one. The definitive test is the *baryonic* TF at $z > 1$ with reliable ALMA gas masses. Direct dark matter searches continue to return null results (LZ Collaboration's 417-day analysis presented December 2025 is the most sensitive WIMP search ever conducted, finding nothing), consistent with the framework's prediction that no particle dark matter exists.

Cluster scales and the Bullet Cluster. Galaxy clusters — the hardest test for any MOND-like theory — are addressed by the interpolation between the Newtonian and deep-MOND regimes. The simple interpolation function $g_{\rm total} = g_B \cdot \nu(g_B/a_0)$ with $\nu(y) = (1 + \sqrt{1 + 4/y})/2$ matches the Coma cluster to $< 1\%$ in velocity (1260 vs 1270 km/s) and reduces the standard MOND mass shortfall from a factor ~ 2 to ~ 1.0–1.5 for other rich clusters — with the residual attributable to undetected warm-hot intergalactic medium (WHIM). This interpolation is indistinguishable from the deep-MOND limit at galaxy scales (differences < 0.07 dex, well within the observed RAR scatter). The Bullet Cluster — where gravitational lensing peaks at the galaxy positions rather than the dominant X-ray gas — is explained by the non-local character of entropy displacement: the boundary entropy relaxation time is $\sim H^{-1} \approx 14$ Gyr, while the collision crossing time is ~ 0.15 Gyr. The dark gravity is frozen at the pre-collision configuration (centered on the galaxies, which defined the potential wells for gigayears), not tracking the recently displaced gas. This reproduces the observed lensing morphology and makes a testable prediction: very old post-collision systems should show gradual relaxation of the dark gravity toward the gas distribution (GR, §7.3). The same thermodynamic averaging explains why the entropy displacement reproduces the CMB acoustic peak pattern: oscillating perturbations have zero net entropy displacement per cycle (the Clausius relation involves *net* heat transfer), so only the growing mode is tracked — providing non-oscillating potential wells identical to CDM in the linear regime.

Neutrino predictions and the JUNO + DESI record. The framework predicts Majorana neutrinos with normal mass ordering and a hierarchical spectrum, with Σm_ν near the oscillation minimum of 0.059 eV. The DESI DR2 + CMB analysis (Elbers et al., March 2025) reports $\Sigma m_\nu < 0.0642$ eV (95% CL), prefers normal ordering, and bounds the lightest neutrino mass at $m_l < 0.023$ eV — every directly comparable measurement matches the OI prediction. The same analysis finds a 3σ tension with the lower oscillation limit assuming ΛCDM, interpreted as "a hint of new physics not necessarily related to neutrinos" — exactly the structural mismatch that the OI dark energy resolves. JUNO first results (November 2025) deliver world-leading precision on

$\sin^2\theta_{12} = 0.3092 \pm 0.0087$, matching the OI prediction $1/3 - 1/(4\pi^2) = 0.3080$ at 0.14σ against the JUNO direct measurement. The post-JUNO global fit of Capozzi et al. (2025) tightens the world average to 0.3085 ± 0.0073, against which the OI prediction matches at 0.07σ. JUNO's measurement also confirms the persistent solar/reactor 1.5σ tension. The neutrino sector and the dark energy sector are coupled in the framework — both follow from the same lattice structure — so the joint observational support is not independent confirmation of separate effects but a single empirical pattern consistent with the framework's central mechanism.

Gauge coupling prediction. The companion paper [SM §6] extends the derivation chain to the gauge coupling strengths. The fermion-induced coupling gives $1/\alpha_0 = 23.25$ at the matching scale $\mu = M_{\rm Pl}$ — a universal value determined by the lattice structure ($N_f = 6$ flavors, $T(R) = 1/2$), not by the specific bijection φ. Combined with non-perturbative gauge self-energy corrections (from pure-gauge Monte Carlo at the induced coupling) and Standard Model renormalization group running, this reproduces all three SM gauge couplings at M_Z: $1/\alpha_1 = 59.00$, $1/\alpha_2 = 29.57$, $1/\alpha_3 = 8.47$ — matching the observed values to $< 0.1\%$.

Independent corroboration of the trace-out mechanism. The framework's central technical move — that integrating out hidden degrees of freedom from a deterministic substrate produces well-defined non-Markovian visible dynamics, which under the Barandes correspondence becomes quantum mechanics — is now established at theorem level in the open systems literature. Brandner (*Phys. Rev. Lett.* **134**, 037101 and companion *Phys. Rev. E* **111**, 014137, January 2025) proves that for autonomous linear evolution equations, integrating out inaccessible degrees of freedom yields well-defined non-Markovian visible dynamics in a controlled weak-memory regime, with explicit error bounds and a convergent perturbation scheme — derived independently for general open systems with no stake in OI being right. Direct experimental demonstrations exist in controlled physical systems: Mehl et al. (PRL 108, 220601, 2012) showed that hidden slow degrees of freedom in a colloidal system produce non-Markovian visible dynamics violating naive Markovian fluctuation theorems; Gröblacher et al. (Nature Communications 6, 7606, 2015) observed non-Markovian Brownian motion in a macroscopic micromechanical oscillator. On the quantum side, Kim (April 2025) shows that monitored quantum systems are formally quantum hidden Markov models with a rigorous correspondence to classical hidden Markov models — the same formal structure as the stochastic-quantum bridge, developed independently in the monitored quantum systems literature. The framework's central technical move is therefore not speculative; it is the standard way physicists now think about open systems with hidden structure, and the mathematical apparatus has been published in the standard literature within the past 12 months.

No competing framework produces all of these from a single definition. The parallel with the cosmological constant dissolution is exact: the 10^{122} discrep-

ancy is the *information compression ratio* of the trace-out, and the ~95% dark sector is the *gravitational occlusion fraction*. Together, they account for the two largest anomalies in modern cosmology as two aspects of a single phenomenon: the cost of observing the universe from within.

Philosophical Lineage

The paper is a physics paper, but its core claims — that observers face irreducible limits, that two irreconcilable descriptions can both be correct, that incompleteness is a structural feature rather than a deficiency — sit at the intersection of some of the oldest debates in philosophy. A systematic mapping against the major traditions reveals a striking pattern: broad support for most of the framework, and near-universal resistance to one specific thesis.

The seven claims

The framework rests on seven implicit philosophical commitments:

1. **Access is specified.** A proper instantaneous readout can omit state distinctions; retained records may recover them.
2. **QM and GR are both correct** within their domains.
3. **Residual inaccessibility is interface-relative.** A hidden quantity remains unidentified when it varies within a compatible record fibre for the permitted preparations, readouts and actions.
4. **The underlying reality is local and definite.** Indeterminacy belongs to the observer's description, not to the world.
5. **Incompleteness has a precise carrier.** Non-identifiability follows from identical permitted records, while computational cost and physical record capacity are separate restrictions. It is not a universal consequence of self-reference.
6. **The description is observer-relative.** Different partitions yield different emergent physics.
7. **Descriptions can be projections of a shared model.** Which distinctions those projections omit depends on their full observation interfaces.

Claims 1, 5, 6, and 7 enjoy broad philosophical support across nearly every tradition examined. Claim 4 — that the underlying reality is definite — is the paper's most philosophically isolated thesis.

The Gödel connection

Gödel's incompleteness theorems concern the provability limits of formal theories under their stated hypotheses. They do not establish that an embedded observer can never infer a finite hidden state. The distinction is concrete: in the known reversible swap model, two retained visible readings identify the complete

initial state. Proposition 24.5 of [Substratum] gives the exact criterion: a quantity is determined by a record when it is constant on the states compatible with that record. Permanent passive ambiguity requires non-singleton future-trace classes on the specified preparation support.

The cited work on physical inference and self-measurement, including Wolpert [19] and T. Breuer (1995, *Philosophy of Science* 62: 197–214), provides comparisons whose own device, measurement and counterfactual hypotheses must be checked. Such results cannot replace a proof about the record interface used here. In particular, a bound on a measuring apparatus's possible internal records can prevent it from distinguishing every admissible total state without making every possible record ambiguous. For a record store with R values and more than R admissible initial states, some record fibre is non-singleton; other fibres may identify the state exactly.

The framework's proved limitations are correspondingly specific. A finite-range dependence graph bounds direct causal influence. The hidden-capacity inequality bounds predictive memory. C4 forces hidden predictive structure relative to the present visible value, while the full retained history may determine that hidden structure. A physical causal horizon restricts direct signals under its spacetime assumptions; it does not alone prohibit inference from accessible correlations. These statements concern specified dynamics, preparations and resources, rather than self-inclusion by itself.

The finite observable-law correspondence $S \iff D \iff Q_{\mathrm{fb}}$ is exact and universal within its stated horizon and fixed-basis readout. It supplies a representation of a law, not a proof that posterior opacity selects quantum mechanics or a unique physical substratum. The simultaneous coherent-instrument completion and physical branch selection remain separate research obligations. Logical incompleteness, finite information loss and physical causal access can therefore be compared without treating one as a derivation of the others. Bell assumptions are separate again: the adopted measurement-independent branch requires ontic parameter dependence with operational no-signaling, as stated in [Main §3.3].

Hofstadter's *Gödel, Escher, Bach* offers a philosophical comparison between microscopic descriptions and emergent levels. Here the exact mathematical result concerns a finite observable law and its equivalent representations. Whether a particular representation supplies the full physical quantum theory, a preferred substratum, or a correct account of vacuum energy remains a separate scientific question. The comparison with a self-referential observer supplies none of those missing proofs and does not make hidden states universally uninferable.

Closest ancestor: Nicholas of Cusa

Of all thinkers surveyed, the fifteenth-century cardinal Nicholas of Cusa provides the most precise structural alignment. His *docta ignorantia* (learned ignorance) is essentially Claim 5: the highest knowledge is knowing what we cannot know,

and this is an intellectual achievement, not a failure. His *coincidentia oppositorum* (coincidence of opposites) maps onto Claim 7 with remarkable precision: contradictions irreconcilable in the finite realm dissolve in infinity — the infinitely large circle's circumference becomes a straight line, the infinite polygon becomes a circle. The framework proposes a common underlying construction for QM and GR; whether the necessary physical bridges hold is a scientific question, not a consequence of this philosophical analogy.

Most strikingly, Cusa's "wall of Paradise" from *De Visione Dei* maps onto Claim 3: an insurmountable boundary beyond which finite intellect cannot pass. Scholars like Emmanuel Falque interpret reaching this wall not as escaping through it but as *inhabiting the boundary* — compatible with a framework where understanding the limit is itself the deepest available insight.

Deepest ontological parallel: Spinoza

Spinoza's attribute theory is arguably the single closest ontological parallel. One substance (God/Nature) expresses itself through infinite attributes, of which humans know only two: Thought and Extension. Jonathan Bennett's "barrier doctrine" captures the key feature: each attribute must be conceived through itself — no explanatory flow crosses between them. This is precisely Claim 7: two complete, correct descriptions that are structurally irreconcilable, yet both describe the same underlying reality. Spinoza's substance is fully determinate, aligning with Claim 4. His parallelism doctrine — the order of ideas is the same as the order of things — means the two descriptions track the same structure through incommensurable vocabularies.

A universal epistemological conflict with Spinoza's *scientia intuitiva* does not follow from the finite results here. Some specified observation interfaces leave genuine unresolved distinctions, while others allow complete state inference from retained records.

The recurring fault line: Claim 4

Across every tradition examined, a striking pattern emerges. The claim that underlying reality is "local and definite" faces resistance from virtually every direction:

Kant prohibits positive characterization of the noumenal realm as dogmatic metaphysics — you can know *that* things-in-themselves exist but never *what* they are. **Hegel** diagnoses a performative contradiction: to posit a hidden sector and characterize it as containing standard physics is already to have crossed the boundary you claim is uncrossable. **Nietzsche** attacks it as residual Platonism — having correctly shown that all observation is perspectival, the paper reinstates the very "true world" his career was dedicated to destroying. **Wittgenstein** rejects it as nonsensical: asserting what lies beyond the limits of the sayable transgresses exactly the limits the paper identifies.

Nagarjuna identifies it as *svabhāva*-reification — attributing inherent existence to what Buddhist emptiness (*śūnyatā*) says lacks it. **Daoism** warns against naming the unnameable: the Dao that can be spoken is not the eternal Dao, and calling reality "definite" is itself an act of conceptual carving. **Whitehead** insists reality is fundamentally processual and creative, involving genuine indeterminacy — removing that indeterminacy robs the cosmos of its creative character.

Comparisons with Advaita Vedanta or other philosophical traditions concern interpretations of access and underlying reality. The mathematical framework does not prove a metaphysical thesis of permanent inaccessibility: its precise limits are non-identifiability relative to specified records and operations. Such comparisons cannot replace that criterion or its physical hypotheses.

Only Spinoza (whose fully determinate substance is naturally definite) and Worrall's epistemic structural realism (which posits real but unknowable natures behind structures) provide genuine philosophical support for Claim 4.

What's genuinely new

The paper's philosophical lineage is not a single line of descent but a mosaic. Its structural epistemology draws from Kant through Wittgenstein to Wolpert. Its complementarity thesis combines Bohr's physics with Cusa's coincidence of opposites and Daoist yin-yang. Its projection metaphysics reaches through Plato's cave and Plotinus's emanation to Advaita Vedanta's maya-Brahman distinction.

The philosophical proposal combines a deterministic model with descriptions relative to specified observer interfaces. Its mathematical content lies in the finite-law representation theorem, the predictive-memory results and explicitly conditional reconstructions. Whether unobserved variables remain unidentified is settled by the permitted record fibres, not by a claim to know the nature of something proved universally unknowable.

Frequently Asked Questions

"Isn't this just another interpretation of quantum mechanics?" The finite-horizon theorem derives an exact stochastic/reversible/fixed-basis quantum representation correspondence. It does not establish a unique physical substratum or the full coherent measurement theory from partial observation alone. The model also develops physical relations under the separately named matter, horizon and completion hypotheses. Those mathematical constructions and open physical bridges are more specific than a philosophical interpretation, while their empirical warrant must be assessed under their stated assumptions.

"Doesn't Bell's theorem rule out hidden variable theories?" Bell's theorem applies directly to the deterministic completion. If its response-complete

hidden state is measurement-independent and parameter-independent, the model is Bell-local and obeys $|S_{\mathrm{CHSH}}| \leq 2$. The Bell mechanism is separate from temporal indivisibility: the framework retains measurement independence, gives up ontic parameter independence for Bell-violating substrata, and separately requires operational no-signaling. The $2\sqrt{2}$ Tsirelson theorem belongs to the weaker stochastic causal-local construction, not to a Bell-local deterministic completion.

"If everything is deterministic underneath, where does randomness come from?" Within a deterministic completion, unresolved initial or hidden variables can give probabilistic reduced predictions. The relevant distribution is conditional on the available record and preparation. Some records identify the state and make its next visible value predictable; others leave a nontrivial distribution. Partial observation alone does not prove irreducible randomness for every embedded observer or every history [Substratum Proposition 24.5].

"What about dark matter?" Dark matter is not a substance — it is the local shape of the observer's blind spot, the same entropy that appears as dark energy when uniform. Baryonic matter displaces boundary entropy; the Jacobson mechanism converts the gradient into curvature; the crossover acceleration $a_0 = cH/6$ and baryonic Tully-Fisher relation $v^4 = GM_B \cdot cH/6$ follow with no fitted parameters, at the G1 register and conditional on the covariant rung G3. The mechanism works across all scales — from galaxies through clusters (Coma matched to $< 1\%$) to the Bullet Cluster and the CMB acoustic peaks — as detailed in the *Predictions* section above.

"What about the Bullet Cluster?" The horizon's response time ($\sim H^{-1} \approx$ 14 Gyr) vastly exceeds the collision crossing time (~ 0.15 Gyr). The boundary entropy is frozen at the pre-collision configuration — centered on the galaxies, not the recently displaced gas. The observed lensing morphology follows without collisionless particles (see *Cluster scales and the Bullet Cluster* above).

"How does this relate to recent mainstream work on observers in physics?" A substantial body of work since 2022 has independently arrived at the foundational claim that observation cannot be treated as external to physics — the same starting point as this framework, reached by different routes. Chandrasekaran, Longo, Penington, and Witten (CLPW, 2022) showed that adding an observer to quantum field theory in a gravitational subregion produces a finite well-defined entropy where previously there was none. Maldacena (2024) showed that an unphysical phase in the de Sitter partition function — long understood to obstruct a clean thermodynamic interpretation of the cosmological horizon — cancels exactly when an observer with a clock is properly incorporated. Harlow, Usatyuk, and Zhao (2025) and the AAIL construction (2025) showed that the dimension of the Hilbert space of quantum gravity in a closed universe is determined by the observer's degrees of freedom rather than being an independent parameter. Slagle and Preskill (2022) explicitly constructed a model in which quantum mechanics emerges at the boundary of a classical lattice with stochastic dynamics in an extra spatial dimension — a logical demonstration

that QM-from-classical is achievable.

The framework here converges with these programs on the foundational substance — observation matters; the algebra of observables depends on the observer; the partition between observer and observed is physically meaningful — while reaching that conclusion through a different route. CLPW, Maldacena, and HUZ all add observer degrees of freedom *to* an existing quantum-gravitational formalism. The framework here goes the other direction: it starts with a deterministic substratum and a partition (the observer and the rest), and derives both quantum mechanics and the gravitational sector as consequences of that structural setup. There is no extra spatial dimension as in Slagle-Preskill, and no fundamental stochasticity. The substratum is finite and deterministic; the observer's incomplete access to it is what produces the appearance of quantum behavior. Among emergent-QM programs the framework is distinguished by carrying twenty-two classified quantitative predictions to data (two parameter-free structural and five conditional structural) — a comparable empirical record is not currently produced by the alternatives. The convergence on observer-essentiality from many independent directions is encouraging context; the framework's empirical record is what makes the case specifically for this realization.

"How does this relate to string theory specifically?" String theory is the dominant unification program in fundamental physics, with several decades of work on whether the Standard Model can be reproduced from compactifications of higher-dimensional string vacua. The framework's relationship to string theory is examined directly in `Structure`. The findings: at the substratum level, the two frameworks are not equivalent. The framework's substratum is a finite deterministic bijection; string theory's matrix-model formulations are continuum stochastic path-integrals with non-linear Yang-Mills interactions. These structural commitments are different in ways that cannot be bridged by simple discretization or limiting procedures.

But the relationship at the observable level is more interesting. A four-feature audit of partial-trace observation — Born rule, channel-level unitarity, memory/divisibility (non-Markovian marginal), commutant gauge-invariance — finds that the first two transport directly between the frameworks and the third at the memory level. Holographic effective field theory in AdS/CFT explicitly classifies modes by Markovianity index, providing a direct analog of the framework's hidden-memory mechanism; P-indivisibility is not established cross-framework. AdS/CFT bulk reconstruction provides a direct analog of the framework's Stinespring dilation, with bulk operators encoded in boundary subregions through quantum-error-correcting-code structure. The shared content is the structural features of partial-trace observation — what observation extracts from any quantum state under partial trace, independent of the substratum-level machinery producing the state. The fourth feature (commutant gauge-invariance) transports as a pattern but not as a specific output: both frameworks produce gauge invariance via commutant restriction,

while the selected OI construction gives an exact cubic decomposition and a conditional condensate stabilizer followed by a specified quotient and charged action [SM Propositions 14a and 15a]. Its physical carrier and action selection, like the compared compactification choices, are additional premises.

`Structure` develops a program for comparing embedded observers through a specified invariant observable algebra and its adjoint channel dynamics. Matching partial-trace features or the SM gauge target is weaker than constructing the required algebra isomorphism and intertwiner. Different model families can share those coarse features while predicting different Cabibbo angles or masses; a carrier containing the measured values places such models in different observational classes. Whether a chosen OI realization and a chosen string construction occupy the same class remains an open comparison problem. The proposed structural-realist interpretation depends on completing that comparison, rather than on declaring their microscopic descriptions interchangeable.

"Why does gravity 'see' the classical vacuum energy and not the quantum one?" Because the spacetime metric exists at the classical level, *before* the quantum description emerges. The quantum zero-point energy is a feature of the observer's compressed description. It's real for quantum experiments (the Casimir effect, the Lamb shift) but doesn't appear in the stress-energy tensor that governs curvature. The 10^{122} discrepancy is the information compression ratio — the entropy of the observer's blind spot.

"Does a deterministic substratum imply permanently inaccessible reality?" No. A definite microscopic state and deterministic dynamics are premises of the chosen model class. The finite-law construction shows that such a completion can represent the specified observable laws; it does not prove that nature has selected that completion. Within a known model, retained records may determine the hidden state, or a non-singleton record fibre may leave it unresolved. Claims about physical inaccessible regions, storage resources and observational equivalence require their own stated hypotheses. Neither permanent opacity nor the actual physical substratum follows from determinism alone.

"Doesn't holographic physics show that spacetime comes from entanglement?" This is probably the strongest objection to the framework's ordering — classical spacetime first, quantum mechanics second. The Ryu-Takayanagi formula says entanglement entropy equals boundary area divided by $4G$. Van Raamsdonk argued that reducing entanglement disconnects spacetime. Programs like ER=EPR and "it from qubit" read these results as evidence that quantum entanglement is prior to geometry.

The framework offers an alternative reading. If the quantum description is produced by tracing out over a geometric boundary, then *of course* its entanglement entropy is proportional to the boundary's area — the information content of the trace-out is set by the number of modes crossing the boundary, which scales with area. The Ryu-Takayanagi formula, on this account, isn't a hint that entanglement builds geometry; it's a consequence of the fact that geometry built

the entanglement. The area law $S = A/(4\,l_p^2)$, which the paper recovers under its horizon calibration, is exactly this statement.

The correlation between entanglement and geometry is real either way. The question is which direction the arrow of explanation points. The two orderings make different empirical predictions: if geometry emerges from entanglement, spacetime structure should break down at high energy; if quantum mechanics emerges from geometry, the quantum description should break down near the discreteness scale while the geometric substratum persists. The cumulative case for the geometry-first ordering: it is consistent with the observed vacuum-energy normalization, it reproduces the Bekenstein-Hawking 1/4 factor as a consistency relation of its horizon calibration, which relates $\hbar$ to the discreteness scale, and it gives the dark-sector concordance, the smooth geometry-dependent dark-energy bookkeeping read in the Type II channel ($\nu \approx 0$, observationally $\approx \Lambda$CDM), the MOND acceleration scale $a_0 = cH/6$, the baryonic Tully-Fisher relation, the SM gauge group with its coupling strengths (the strengths through three matched parameters), the Coma cluster mass, and the Bullet Cluster lensing morphology — five of these mutually independent comparisons with data, the MOND scale carrying an imported factor ([GR §6.2]). The quantum-first ordering produces the 10^{122} discrepancy, treats $\hbar$ as an unexplained input, and has no natural account of the dark sector.

The Decompression Algorithm

There is a way to state what the framework means that is sharper than anything in the formal paper.

An observer inside the universe receives incomplete data — the visible sector only. On every finite horizon those statistics, Markov laws included, have an exact fixed-basis quantum representation — a unitary system measured projectively by the Born rule in one fixed basis — so a fixed-basis unitary/Born algorithm reproduces those statistics whatever the law ([Main §3.4]). The statistics carry positive conditional memory $I(X_{<k}; X_{k+1} \mid X_k)$ at an accessible order k exactly when history readback holds at that order ([Main §3.4]). Under the translation hypothesis (T), processes in the correspondence's class admit the compressed form of the representation; identifying one standard quantum theory for all coherent interventions and composites, the algorithm whose features follow, requires the operational-lifting and composition hypotheses of [Main §3.4].

Granted that identification, everything in QM is a feature of that algorithm, not a feature of the underlying reality. The wave function is the algorithm's internal state variable — the bookkeeping device it uses to track what it knows and what it doesn't. Complex amplitudes are the algorithm's arithmetic — the specific number system the reconstruction requires. Interference is what happens when the algorithm combines two incomplete pathways and their bookkeeping entries partially cancel. Entanglement is the algorithm's encoding of correlations that

were written into the hidden sector during preparation and haven't been read back yet. The Born rule is the algorithm's output format — the way it converts its internal state into predictions the observer can check.

None of these are properties of the deterministic substratum. In the substratum, there are no wave functions, no complex numbers, no interference, no entanglement. There are configurations evolving under a Hamiltonian flow. That's all. The entire apparatus of quantum mechanics — every textbook, every equation, every experiment — is what the decompression algorithm looks like from the inside.

Antimatter as algorithmic artifact

This reframing changes the understanding of specific phenomena. Consider antimatter.

The standard account: the Dirac equation, which combines quantum mechanics with special relativity, has solutions with both positive and negative energy. The negative-energy solutions correspond to antiparticles — particles with opposite charge and quantum numbers. Every particle must have an antiparticle, because CPT invariance (a theorem of any local Lorentz-invariant quantum field theory) requires it.

The framework's account: the trace-out over the hidden sector forces the reconstruction algorithm to use two-signed amplitudes. When the algorithm operates in a relativistic context — which it must, because the substratum's causal structure is relativistic — the two-signed amplitude structure becomes the two-signed energy structure of the Dirac equation. Negative-energy solutions aren't additional features of reality. They're what the algorithm requires for self-consistency when the incomplete data comes from a relativistic system.

The parallel to the coin-and-die model is direct. The intermediate propagator:

$$\Lambda(2,1) = \begin{pmatrix} 2 & -1 \\ -1 & 2 \end{pmatrix}$$

has entries of -1 — "anti-probabilities" that don't exist in the substratum (where every transition probability is between 0 and 1). These negative entries are the minimal-model ancestors of antimatter. They arise for the same reason: the algorithm can't reconstruct the observed dynamics without them. In the toy model, the -1 entries encode the fact that the die remembers and reverses the coin flip — information backflow that no positive-entry matrix can describe. In the relativistic case, the negative-energy solutions encode the fact that the relativistic trace-out requires a doubled Hilbert space to remain self-consistent.

In both cases: the substratum has no doubling. The algorithm demands one.

Nothing in quantum mechanics explains a fundamental phenomenon

This is the deepest implication of the framework, and it is worth stating plainly.

The wave function does not explain what an electron is doing between measurements. It encodes what the algorithm computes given the observer's data. Interference does not explain why particles behave like waves. It reflects the algorithm's method of combining incomplete information. Entanglement does not explain a mysterious connection between distant particles. It reflects correlations stored in the hidden sector that the algorithm must track but cannot directly access. Antimatter does not explain a second kind of fundamental stuff. It reflects the algorithm's need for two-signed amplitudes in a relativistic context.

The finite-law theorem gives an exact fixed-basis quantum representation of the accessible stochastic law. It does not select a unique microscopic realization, and equality with the full coherent instrument/composite theory retains the separate operational-lifting conditions. The decompression analogy describes this representation, not a theorem that one complete physical ontology is forced.

This doesn't make quantum phenomena less real. Temperature is also an emergent phenomenon — a single molecule has no temperature. But temperature boils water, drives engines, and burns skin. Its emergence from statistical mechanics doesn't diminish its causal power within the emergent description. The same holds for every quantum phenomenon: they are causally potent, experimentally real, and the only physics accessible to an embedded observer. But they are features of the observer's necessary algorithm, not features of the universe's fundamental structure.

This resolves what is arguably the oldest open question in quantum foundations: *is the wave function real?* Since 1926, physics has been split between epistemic interpretations (the wave function is just a bookkeeping device tracking the observer's knowledge — it doesn't correspond to anything physical) and ontic interpretations (the wave function is a real physical entity — a field on configuration space, or a branching structure, or a guiding wave). The framework shows that both sides share a hidden assumption: that "real" means "fundamental." The epistemic camp says the wave function isn't fundamental, therefore it isn't real. The ontic camp says it's real, therefore it must be fundamental. Both are wrong. The wave function is real *and* not fundamental — in exactly the same way as antimatter, temperature, and every other emergent phenomenon. Within the bounded projection — which is all an embedded observer will ever have access to — the wave function, the positron, the interference pattern, and the electromagnetic field all have exactly the same ontological status. They are all outputs of the decompression algorithm. They are all experimentally real. And none of them exist in the substratum.

The analogy to data compression is precise. Lossy compression (like JPEG or MP3) discards information and reconstructs an approximation. The reconstruc-

tion has artifacts — ringing near edges in JPEG, pre-echo before transients in MP3. These artifacts are not in the original signal. They are features of the reconstruction algorithm applied to incomplete data.

Quantum mechanics is the reconstruction. The 10^{122} discarded bits (the hidden sector) are the lost information. Wave functions, interference, entanglement, and antimatter are the artifacts. They are real — as real as JPEG ringing is visible — but they are properties of the reconstruction, not of the original.

The compression analogy has a limit. Observable data can admit multiple reversible realizations and quantum representations. The exact finite-horizon correspondence establishes existence in both directions, not one unique reconstruction algorithm. Recovering a physical local branch and a full operational theory requires the additional selection and compatibility conditions discussed in the technical papers.

What This Means

The search for a "theory of everything" that unifies quantum mechanics and general relativity has assumed that the two theories describe the same level of reality and must be reconciled there. This paper argues that assumption is wrong. The two theories operate at different levels — one fundamental, one emergent — and their apparent contradiction is the information-theoretic cost of being an observer trapped inside the system you're trying to describe.

The apparent conflict between QM and GR, the 10^{122} vacuum energy discrepancy, and the dark sector are three faces of the same fact. The first is the existence of the trace-out. The second is its information compression ratio. The third is its gravitational occlusion fraction. All three are mandatory consequences of observational incompleteness — and all three match what we observe.

The universe is not broken. We are observing it from within.

Corroboration: The Rigidity Test

A natural objection to any framework this sweeping is: maybe it only works because it was built to work. Maybe the definition was chosen to produce QM, and the cosmological application was chosen because it fits. A flexible framework that can accommodate anything predicts nothing.

The companion papers `Substratum` and `SM` test this by asking a question the foundational paper doesn't address: if you build a concrete system satisfying the definition, does the framework constrain the dynamics? And if so, does the constrained dynamics produce anything beyond QM — something the framework wasn't designed to deliver?

It does, on both counts.

The dynamics is unique. Among all second-order reversible nearest-neighbor dynamics on a lattice, the requirements of center independence (necessary for emergent QM), spatial isotropy, and linearity select exactly one: the discrete wave equation. This holds for any alphabet size and any dimension. Center independence is necessary because center-dependent rules allow the visible sector to partially predict itself, suppressing the information backflow that produces the quantum phenomenology — an effect proven analytically on the full lattice via an information-screening mechanism. Linearity is selected by three independent criteria: it gives maximum propagation speed, it uniquely maximizes P-indivisibility among linear rules, and it is the only choice for which horizons are in thermal equilibrium (nonlinear dispersion breaks the Unruh effect).

The dynamics passes seven independent checks. The wave equation — selected solely by the QM requirement plus symmetry — turns out to produce: (1) non-Markovian reduced dynamics (emergent QM), (2) a causal structure with the correct spacetime dimension, (3) the gap equation for $\hbar$, (4) entanglement entropy proportional to boundary area, (5) a Lorentz-invariant dispersion relation at leading order, with linear Lorentz-invariance violation forbidden at the substrate level (the cubic-lattice dispersion admits no $\mathcal{O}(k^3)$ term) and the leading correction fixed at $\omega^2 = k^2[1 - (4\sigma/45)(E/M_{\rm Pl})^2 + ...]$, a coefficient $4/45$ at one nat of entropy per boundary mode — subluminal, with the lattice coefficient $1/45$ of $(k\epsilon)^2$ a specific prediction, not a free parameter, while the amplitude $4\sigma/45$ carries the undetermined σ, so a measurement of it would constrain σ, (6) the correct Unruh temperature at horizons, and (7) all inputs to Jacobson's thermodynamic derivation of Einstein's field equations. Each of these is an independent check that could have come out wrong — the entropy could have scaled with volume, the dispersion could have been non-relativistic or admitted linear LIV, the horizon state could have been non-thermal. None of these failures occurs.

No free parameters. The wave equation is selected, not chosen. The lattice spacing is fixed by the gap equation. The entropy coefficient is determined by thermal matching. The dispersion relation is an algebraic identity. There is nothing to tune.

This is what rigidity looks like. A framework that produces the quantum description from one set of arguments and then, without modification, produces the inputs for GR from a completely different set of arguments — lattice dynamics, dispersion relations, entanglement Hamiltonians — is making a structural claim that goes well beyond the original derivation. Each passed check is a point where the framework could have been falsified and wasn't. Seven passed checks with no free parameters is not proof, but it is the kind of evidence that distinguishes a framework describing something real from one that was engineered to fit.

What Is the Lattice?

The rigidity test proves that the wave equation on a lattice produces both QM and GR. But what *is* the lattice? Is the universe literally a grid of cells at the Planck scale? What would the grid be made of? What would it sit in?

The companion papers `Substratum` and `SM` address these questions head-on — and also derive the Standard Model's structure from the lattice dynamics. Their answer to the ontology question: the lattice is not a physical object. It is the *coupling structure* of the dynamics — the pattern of which degrees of freedom affect which others.

The minimal structure. The paper audits the assumptions its theorems consume and identifies which are necessary for the theorems and which are artifacts of the particular construction. The result: on a state space identified with a product of local degrees of freedom — site factorization, itself an input that the dynamics has not been shown to select — six structural properties carry the emergent-QM, dimensional, and gravitational content: deterministic bijectivity, finite boundary entropy, bounded coupling degree, statistical isotropy, non-trivial partition coupling, and slow-bath capacity. That content does not use the specific alphabet size q or the wave equation; the wave equation's form rests on further assumptions, center independence, linearity and the nearest-neighbour range. The *Standard Model gauge group and the quantitative predictions* are the exception: they additionally require the simple-cubic Bravais structure (the multiplicities (3,2,1) come from the octahedral point group), which is an empirically-selected input forced by the observed SU(3) factor, not derivable from statistical isotropy alone. For the emergent-QM core the minimal object is not a lattice; for the Standard Model it is the cubic lattice specifically. It is a triple: a finite set S, a bijection φ, and a partition V.

Space as coupling graph. Given any bijection on a finite set, you can ask: which components of the state affect which others? The answer defines a graph — sites are vertices, and two sites are connected if the dynamics couples them. This graph is not drawn on a pre-existing space. It IS space, at the fundamental level. Distance means "how many coupling steps apart." Area means "how many edges cross the boundary." The area law, the dispersion relation, the Myrheim-Meyer dimension — every spatial property used in the derivation chain is a property of this coupling graph.

This dissolves the container problem. The lattice doesn't sit in anything. There is no ambient manifold. The coupling graph is an abstract mathematical structure — like a number or a group — and any spatial embedding we draw is a representation for our convenience, not a physical fact. Asking "what does the lattice sit in?" is like asking what the number 7 sits in. It's a category error.

The alphabet size. Several results hold for every alphabet size $q \geq 2$ and contain no q: P-indivisibility, center independence, the boundary scaling of the area law, and the dispersion relation of the observer-level operator. Others

carry q: the period of the wave map, the coefficient of the area law, the capacity count of the hidden sector. In the gravitational sector q could enter only through the entropy assigned to one boundary cell, which cancels from the Bekenstein-Hawking formula. That no observable depends on q at all is a hypothesis, the alphabet-size hypothesis [SM §2.7]. Under it q is a gauge freedom — a choice of mathematical description, like choosing a coordinate system or a gauge for the electromagnetic potential — and the physical content is the coupling structure, not the microscopic state space.

Connection to causal set theory. The bijection's coupling graph, extended in time, generates a causal partial order: event A precedes event B if B is within A's future coupling light cone. This partial order is a causal set in the sense of Bombelli, Lee, Meyer, and Sorkin — the starting point of causal set theory, one of the established approaches to quantum gravity. Causal set theory has always had a specific gap: it postulates a causal order but lacks a deterministic dynamics yielding the quantum and gravitational descriptions. The OI framework provides exactly this. Sorkin's slogan is "Order + Number = Geometry." The OI version is "Bijection + Locality = QM + GR."

The hierarchy of physics. If the fundamental object is (S, φ) — a finite set and a bijection — then every concept in physics is a different aspect of this pair:

- φ is the dynamical law — the complete rule mapping states to states.
- **Space** is the coupling structure of φ — which degrees of freedom affect which others. It is determined by φ as the factorization minimizing coupling degree. Space is not a container; it is a relationship.
- **Matter** is the state — the values assigned to the degrees of freedom. A particle is a localized pattern that propagates through the coupling graph. Space and matter are the graph topology and the graph coloring; both come from (S, φ).
- **Energy** is the rate of change of the state under iteration of φ. A high-energy excitation changes rapidly from step to step; the vacuum changes least. Energy is not a substance — it is a measure of how fast a pattern evolves.
- **Time** is the iteration itself — the stepping from one state to the next. There is no continuous time at the fundamental level.
- **Quantum mechanics** is the observer's compressed view of V. It exists because the observer must marginalize over the hidden sector.
- **Gravity** is the thermodynamic limit of the coupling structure — the macroscopic behavior of bounded-degree graphs with area-law entropy.
- **Conservation laws** are emergent. Bijectivity of φ preserves state-space volume (information conservation). Energy conservation is what this looks like in the emergent quantum description (Noether's theorem). Momentum conservation reflects the symmetry of the coupling graph.

None of these are independent substances. They are the same structure viewed at different scales or from different angles. The framework does not unify them by reducing them to a common material. It unifies them by showing they were

never separate.

What remains. The structural reading leaves little genuinely open. The spatial dimensionality $d = 3$ is derived by three independent self-consistency filters (propagating gravity, stable matter, and the boundary entropy concordance), with renormalizability of the emergent Yang-Mills coupling as a downstream consequence rather than an independent filter. Background independence is achieved through state-dependent bijections, with the discrete Einstein equation identified as the Ollivier-Ricci curvature condition. The observer is proved to be generic: any small subgraph of any large bounded-degree bijection satisfies the conditions for emergent QM. The fundamental object is (S, φ) — a finite set and a bijection. The partition V, the dimension, and the laws of physics are all derived. What remains undetermined are the 18+ parameter values of the Standard Model, which depend on the specific bijection φ — analogous to how Einstein's equations don't determine the mass of Jupiter.

But the central claim stands: the fundamental object is (S, φ) — a finite set and a bijection. The observer, the dimension, and the laws of physics are all emergent.

Why These Particles?

The first two companion papers establish the framework's identification of quantum mechanics with embedded observation and its derivation of general relativity. But quantum mechanics is a *framework*, not a *theory*. It tells the observer to use Hilbert spaces, unitary evolution, and the Born rule — but it's compatible with infinitely many different quantum field theories. You could have quantum mechanics with an SU(7) gauge group, or with 15 generations of fermions, or with no gauge fields at all. Identifying QM with embedded observation answers the question "what kind of probability does the observer use?" It doesn't answer "what particles exist?" or "what forces act between them?"

The Standard Model of particle physics — quarks, leptons, the strong and weak nuclear forces, electromagnetism, the Higgs boson — has been confirmed to extraordinary precision. But its *structure* has always been taken as empirical. We observe three generations, SU(3) × SU(2) × U(1), specific hypercharge assignments, and we accept them as given. Nobody has explained *why* these particles and not others.

`SM` asks whether the specific lattice dynamics selected by the QM and GR requirements — the wave equation on a d = 3 hypercubic lattice with checkerboard partition — determines which quantum field theory the observer sees. The answer is yes — almost completely.

Fermions from the wave equation. The wave equation selected by the QM requirement (the discrete lattice Klein-Gordon equation) has a well-known mathematical property: it factors into first-order operators called staggered

Dirac operators. This factorization, discovered by Susskind in 1977, means the *same dynamics* that produces bosonic waves also describes fermionic matter. Fermions are not added to the framework. They *are* the framework, seen from a different mathematical angle.

The factorization produces a specific structure: on a three-dimensional lattice, the staggered construction yields exactly 4 "tastes" — independent species of fermions. These decompose as $1 + 3$ under the cubic symmetry of the lattice: one singlet and one triplet. The triplet count equals the spatial dimension $d = 3$. Since $d = 3$ is derived in `Substratum`, the three-generation structure of the Standard Model — the fact that there are three copies of each type of quark and lepton — is traced back to the dimensionality of space.

Numerically, the singlet and triplet are not just distinguished by symmetry — they have quantitatively different coupling strengths. The singlet has coupling $|\mu| = 1$ (the maximum), while the three triplet members each have $|\mu| = 1/3$. The triplet members are exactly degenerate by cubic symmetry, providing a lattice-level origin for "generation symmetry."

Grading and the separate Higgs construction. The recovered [SM Theorem 3] is an exact first-order statement: over a coefficient ring with no additive 2-torsion, a site-diagonal term added to fixed odd hopping preserves the grading precisely when that onsite term vanishes. It does not identify a second-order wave-equation center with a physical chiral mass, or derive spin-chiral fermions and a Higgs from center independence alone. Main's hidden-memory results have their own hypotheses; physical spin and weak chirality retain H-spin'/H-χ'; and the Higgs interpretation and charged matter action require their separate construction.

Gauge structure from a matrix. The wave equation naturally generalizes to multiple components. Each lattice site carries a K-component vector instead of a single number. The selection criteria (center independence, isotropy, reversibility, linearity) uniquely produce the *matrix wave equation*, where a coupling matrix M is the sole new parameter.

At the symmetric nearest-neighbor point the coupling matrix is scalar, so its unitary commutant is the full $U(6)$. Under H-link single-copy and H-cust, a specified equivariant condensate Σ instead carries the three eigenspaces used in the block calculation. Its distinct eigenvalue multiplicities determine its exact unitary stabilizer. Physical masses, the charged observer action and the selected gauge quotient require the further premises stated in [SM §§4.4–4.9].

Why K = 6 on the selected branch. A three-dimensional simple-cubic site has six signed nearest-neighbor directions. H-link identifies the physical carrier with the link representation tensored with a multiplicity space, giving $K = 2dm$. The additional single-copy clause $m = 1$ gives $K = 6$. Neither the link count nor coupling-degree minimization establishes that clause or uniquely selects the physical model.

These 6 link directions form a representation of the cubic rotation group O. Standard character theory decomposes this representation into irreducibles:

$$6 = T_1(3) \oplus E(2) \oplus A_1(1)$$

The six-link representation has dimensions 3, 2 and 1. A generic equivariant Hermitian condensate has $\Sigma = \mathrm{diag}(s_c I_3, s_w I_2, s_y)$ with distinct eigenvalues. Its unitary stabilizer is

$$G_0 = U(3) \times U(2) \times U(1).$$

This stabilizer has three central phase directions. The constructive correlated quotient of [SM Proposition 14a] is $S(U(3) \times U(2))$; the specified SM charged family has faithful image $(SU(3) \times SU(2) \times U(1))/\mathbb{Z}_6$ [SM Proposition 15a]. Its Lie algebra is the familiar SM algebra. Choosing the additional redundant central direction and that charged action as physical is not a consequence of the six-link decomposition alone. The cubic blocks supply the nonabelian dimensions; physical carrier, chirality and light-spectrum selection retain their separate hypotheses.

Chiral coupling. The Standard Model has a peculiar feature: the weak force treats left-handed and right-handed particles differently. Left-handed particles form SU(2) doublets; right-handed particles are SU(2) singlets. No other known force has this asymmetry. Where does it come from?

In the OI framework, part of the answer is the partition itself — and part remains open, and is marked as such. The checkerboard partition that produces the memory-bearing dynamics quantum mechanics represents — even sites visible, odd sites hidden — coincides with the natural $\mathbb{Z}_2$ grading of staggered fermions. That grading, however, is the *product* of spin chirality and taste chirality ($\gamma_5 \otimes \xi_5$), not spin chirality alone: each sublattice carries both left- and right-handed components. What the partition does supply is a striking identity: on the visible sector, spin chirality and taste chirality become the same operator. The weak force's handedness then reduces to one sharp question — does the SU(2) embedding select a taste chirality? The minimal link-current embedding is now certified *not* to — it is taste-blind, exactly like color's (`hchi_selectivity_probes.py`, exact) — so the framework records the sharpened hypothesis H-χ': whether a non-minimal mechanism (a taste-asymmetric condensate structure entering the effective theory) restores selectivity. The honest summary: the substratum explains why a chirality-shaped grading exists, certifies that both the color and the minimal weak embeddings are blind to it, and leaves the weak force's specific handedness to the sharpened open mechanism.

Meanwhile, color SU(3) acts on the *internal* components (the K-component index at each site), which are preserved by the spatial trace-out. Both left

and right fields carry color. SU(3) remains vector-like. The distinction: the grading lives in the spatial structure (touched by the trace-out); color lives in the internal structure (unaffected).

The strong CP problem, open. QCD has a parameter $\bar{\theta}$ that violates CP symmetry. Experimentally, $|\bar{\theta}| < 10^{-10}$. Why is it so small? This is the strong CP problem. The standard solution invokes a new particle (the axion).

What the OI framework contributes is a statement of what time-reversal symmetry of the substratum does and does not fix. T-invariance of the substratum wave equation narrows the QCD angle to $\theta \in \{0, \pi\}$ — both are time-reversal-invariant points — and it makes the visible-sector transition counts reciprocal ($T_{ij} = T_{\theta j, \theta i}$, proved and verified numerically). It does not make the emergent Hamiltonian T-invariant: reciprocal transition probabilities can hide a loop phase that no rephasing removes, so the Yukawa determinant phase, and with it $\bar{\theta} = \theta + \arg\det(Y_u Y_d)$, is not fixed ([SM §5]). Strong CP is therefore open under two named conditions — selecting $\theta = 0$ over π, and a mechanism fixing $\arg\det(Y_u Y_d) = 0$ without making the Yukawas simultaneously real, which would also remove the observed CKM CP violation. The framework makes no prediction for the neutron electric dipole moment.

Conditional charge and chirality results. The selected cubic carrier, generic condensate stabilizer, correlated quotient and charged-family image give exact mathematical results under the premises of [SM §§4.4–4.9]. They do not independently select the physical observer action or light chiral family. On the five-representation family of Theorem 15, with no additional charged anomaly cancelers, the nonzero quark-doublet charge branch has the SM ratios up to normalization and quark-singlet interchange; a separate neutral-Q branch also exists. The stated electromagnetic or Higgs/Yukawa input fixes the numerical normalization. Physical chirality retains H-χ', and the complete geometric and dynamical selection problem remains open.

Where the matter representations come from. A subtle point: the K = 6 internal space decomposes as a *direct sum* 3 + 2 + 1, but the SM quark doublet $\mathrm{Q}_L = (3, 2, +1/6)$ is a *tensor product* representation — it transforms simultaneously under both SU(3) and SU(2). This representation doesn't come from the lattice geometry. It comes from the *observer's existence*: the observer genericity theorem ([SM §8.2] Theorem 22) guarantees coupled, capacity-floored partitions generically, and every such observer's finite observable law has the universal fixed-basis quantum representation ([Main §3.4]), internal, with its compressed form under (T). The argument takes the observer's description to be a self-consistent chiral QFT, and a self-consistent chiral QFT requires anomaly cancellation. Anomaly cancellation with SU(3) × SU(2) × U(1) uniquely determines the matter content per generation: Q_L, u_R, d_R, L_L, e_R with specific hypercharges. The lattice gives the gauge group and the generation count; the observer's self-consistency gives the representations. Three degenerate tastes then give three copies — three candidate matter sectors, which read as the three generations under H-spin' ([SM §4.7]).

What the framework does not determine. The coupling matrix M has three independent eigenvalues (μ_c, μ_w, μ_y); the argument that a maximum-speed condition fixes $\mu_y = 1$ is not established (SM §4.6), so all three are solution-level, the first two setting the strong and weak coupling strengths at the lattice scale. These, along with the Yukawa couplings (which determine fermion masses and mixing angles) and the Higgs potential parameters, are properties of the specific bijection φ — the specific "universe" the observer inhabits. The companion papers determine the *form* of the laws but not the *constants*. However, the universal induced coupling $1/\alpha_0 = 23.25$ IS structural — it depends on $N_f = 6$ and $T(R) = 1/2$, both determined by the lattice, not on the eigenvalues of M. The value is the ambient staggered-background coefficient; for the hypercharge factor its reading as a native coupling carries H-observer-bundle and H-Y-vertex ([SM §6.5]), which the non-abelian factors do not inherit. Combined with non-perturbative gauge self-energy corrections and SM RG running, this reproduces all three observed gauge couplings at M_Z ([SM §6]). [SM §7] extends this further: the $d = 3$ lattice geometry organizes twenty-two SM observables — fermion mass ratios, CKM and PMNS mixing angles, the Higgs mass — of which two are parameter-free structural and five conditional structural retrodictions and the remainder are layered, chained, or fitted, each classified explicitly in [SM §7.6].

Predictions. The companion papers make seven structural predictions: (1) no additional gauge groups below the Planck scale; (2) no fundamental scalars beyond the Higgs; (3) no supersymmetric partners; (4) no fourth generation; (5) strong CP open: T-invariance fixes $\theta \in \{0, \pi\}$ and reciprocal transition counts, not $\bar{\theta}$ [SM §5]; (6) flavor-sector CP phases in CKM and PMNS are solution-specific inputs (set by the basis-alignment of the specific φ) — compatible with but not derived by the framework, on the same footing as the fermion mass spectrum [SM §5.4]; (7) neutrinos are Majorana — the singlet taste is the Higgs sector (not a right-handed neutrino), so neutrinoless double beta decay should be observed. The taste-breaking mechanism ([SM §8.4]) further predicts: normal mass ordering ($m_3 > m_2 > m_1$), large PMNS mixing angles (from the structural mismatch between Dirac and Majorana mass matrices under the cubic group), and a hierarchical spectrum with Σm_ν near the oscillation minimum (≈ 0.059 eV). DESI DR2+CMB data ($\Sigma m_\nu < 0.064$ eV, normal ordering preferred) are consistent with all of these.

Combined with the earlier papers' predictions — dark energy in the Type II running-vacuum channel (the functional form is the local expansion shared by every smooth model, [GR §7.1]; the content is the channel — a structural reading, not a derived law — and the magnitude $|\nu| \approx |\nu_{\mathrm{QFT}}| \sim 10^{-32}$, consistent with Bertini 2025's value-consistent-with-zero measurement at 2σ), dark matter from galaxies through clusters to the Bullet Cluster, gauge couplings at M_Z — and [SM §7]'s twenty-two classified quantitative retrodictions (two parameter-free structural and five conditional structural; including $\sin^2\theta_{12} = 1/3 - 1/(4\pi^2)$ matching the JUNO-era global fit at 0.07σ — a retrodiction), the framework makes a suite of falsifiable predictions that no competing theory makes jointly.

The Full Picture

The framework now covers the three pillars of modern physics and their consequences:

Pillar	What is derived	Source paper
Quantum mechanics	Emerges from embedded observation	Main
General relativity	Emerges from lattice dynamics passing 7 checks	GR + Substratum
Standard Model structure	SU(3) × SU(2) × U(1), 3 generations, Higgs, chiral coupling	SM
Structural foundations	$d = 3$, coupling graph ontology, alphabet-size hypothesis, background independence (state-dependent geometry)	Substratum
Chemistry → life	Orbitals, periodic table, carbon, water, chirality, autocatalysis, origin of life, evolution	Complexity

The starting point for all of it: a finite set, a bijection, and a partition.

From physics to life. The derived structure supplies the preconditions for organic chemistry with no fitted parameters: $d = 3$ determines orbital structure, the periodic table, carbon's tetrahedral bonding, water's solvent properties, and a thermal window for dynamic chemistry. The partition produces parity violation, which selects L-amino acids through autocatalytic amplification of the parity-violating energy difference. Combinatorial diversity exceeds Kauffman's autocatalytic threshold by 10^{56}. The origin of life is identified as the first molecular system satisfying C1–C4 — the same conditions that diagnose an accessible non-Markovian realization at the cosmological scale. RNA is the unique molecule that is simultaneously template (C2, C3) and catalyst (C1), making the RNA world a structural prediction rather than a historical hypothesis. C1–C4 systems grow exponentially while ordinary chemistry grows linearly, so the transition is inevitable and irreversible. Darwinian evolution follows from imperfect template replication. The viable parameter fraction is ~16%; fine-tuning and the anthropic principle dissolve.

The framework does not claim to be a complete theory of everything. It does not determine the 18+ parameters of the Standard Model — these depend on the specific bijection φ, just as the mass distribution of the universe depends

on initial conditions, not on Einstein's equations. What it does claim — and what the derivation chain supports — is that the *structure* of the known laws of physics is not arbitrary. It is what mathematical consistency looks like when a finite deterministic system is observed from within. The fundamental object is (S, φ): a finite set and a bijection. Everything else — the observer, the dimension, quantum mechanics, general relativity, and the Standard Model — is emergent.

What Is (S, φ)?

The papers derive what (S, φ) *produces*. But what kind of thing *is* it?

Storage and memory

S is the set of all distinguishable states. Its physical meaning is *finite capacity*: there are |S| configurations that can be told apart, carrying $\log_2|S|$ bits of information. The finiteness is not a regularization — it's load-bearing. Finite sets recur ($\varphi^{\frown}N = \mathrm{id}$), recurrence, together with a non-permutation rooted map, produces P-indivisibility, and the imported correspondence supplies the quantum representation — for its whole class, with P-indivisibility fixing whether the represented dynamics is nontrivially indivisible. Strictly, only the visible sector and boundary layer need to be finite — the deep hidden sector may be infinite, because the readback lemma (Main, §2.3) establishes conditional memory without recurrence (P-indivisibility, which the recurrence theorem supplies, uses the finiteness of the whole state space), and the boundary-only dependence lemma (GR, §3.2) ensures the deep sector's contribution is suppressed by $\tau_S/\tau_B \sim 10^{-32}$.

φ is a bijection: every state has exactly one predecessor and exactly one successor. Its physical meaning is *perfect memory*. Information is never created, never destroyed. The past is always recoverable from the present, because φ^{-1} exists. A non-bijective map would send two states to the same successor, erasing the distinction between them. φ preserves all distinctions, permanently.

Together, (S, φ) is a *finite lossless memory*: a register with finite capacity whose contents are updated without loss at each step. The partition V defines the observer's *access* — both read and write. The observer reads the visible sector (measuring x) and writes to the hidden sector (each visible-sector operation imprints correlations on H through coupling). The hidden sector stores those writes (C2: memory persistence — the record survives the window, whether by slow evolution or by conserved fast dynamics) and returns them on subsequent reads (history readback) — the memory-bearing sector where the distinctly quantum phenomenology lives; representability itself is universal.

The memory interpretation concerns predictive information relative to a specified readout. C1 couples visible coordinates to coordinates not directly

read; those coordinates may nevertheless be inferred from retained records. C4 gives histories with different next-step conditional laws despite the same visible present. C2 concerns persistence to the relevant readback. C3 bounds task-relevant conditional predictive information by the hidden capacity; it does not require storing an uncompressed complete interaction history or imply residual uncertainty after that history is retained.

At the finite observable-law layer, the fixed-basis unitary representation and quadratic Born dictionary are exact ($S \iff D \iff Q_{\mathrm{fb}}$, [Main §3.4]). The Born dictionary is part of that representation's definition; it does not independently select the exponent. The coherent intervention/composite family and a physical measurement mechanism remain separate completion questions. The proposed cosmological storage interpretation, including horizon entropy, vacuum energy and the action scale, depends on the additional physical assumptions of the horizon construction. Those physical claims do not follow from the finite record-fibre or predictive-capacity theorem alone.

An immediate objection is "Memory made of what?" Here (S, φ) specifies a candidate mathematical model, whose accessible data also depend on the observer partition and preparation/readout rule. Theorem 23 gives conditional local outputs after a cubic branch and additional physical hypotheses are chosen. It does not prove that this model exhausts reality or that every possible deeper distinction is inaccessible. Two interpretations stipulated to give identical data cannot be distinguished using only those data; a universal microscopic gauge classification is a separate, open question.

Computation from within

A Turing machine has a tape (storage), a head (read/write access), and a transition function (update rule). Under effective finiteness — where the visible sector and boundary layer are finite but the deep hidden sector may be infinite (Main, §4.3) — the correspondence between (S, φ, V) and a reversible Turing machine is not an analogy. It is structural.

The visible sector V is the head: a finite-state subsystem with bounded local coupling. The hidden sector H is the tape: a memory register that stores correlations written by past interactions. φ is a reversible transition function: no erasure, no halting, no information loss. The conditions C1–C4 characterize this architecture: C1 (coupling) is the read/write mechanism, and C4 (readback) closes the loop: history-sensitive hidden mediation, realized cosmologically as a read-write cycle. C2 (memory persistence) says the tape retains its contents between head operations. C3 (capacity) says the tape is large enough to store the full interaction history.

So what's different? Not the finiteness — the tape can be infinite. Not the reversibility — that's a specialization (reversible Turing machines are a well-studied subclass, going back to Bennett and Fredkin-Toffoli). The one genuine difference is the *question asked*.

Turing asked: what can a machine compute for an external observer who feeds in input and reads output?

The framework asks: what does computation look like to a component of the machine that cannot read the full tape?

At the finite observable-law layer, the answer is an exact quantum **representation**. The head cannot access the tape's global state. It sees only local statistics — transition probabilities obtained by marginalizing over unread tape cells. With readback those statistics are non-Markovian. The fixed-basis Hilbert/unitary/Born-form description is exact ($S \iff D \iff Q_{\mathrm{fb}}$). The stronger statement that superposition, entanglement, and arbitrary coherent instruments all belong to one standard local quantum operational theory is the separate coherent/composite lifting problem, not something the tape analogy proves by itself.

People ask: "Is the universe a computer?" The framework's answer: the universe is a reversible computation. And physics — quantum mechanics, general relativity, the Standard Model — is what that computation looks like to a subsystem embedded within it, observing through a finite window.

The arrow of time

If the universe is a finite permutation, it must eventually cycle back: $\varphi^{\wedge}\mathrm{N} = \mathrm{id}$ for some N. So where does the arrow of time come from? Why does entropy increase?

The substratum itself has no arrow of time. φ and φ^{-1} are equally valid. The total fine-grained entropy is exactly conserved. So the arrow, if there is one, has to come from somewhere else — specifically, from the observer.

A structural theorem (Main §4.6) establishes exactly that, under a stated mixing hypothesis for equilibrium regions and for the coarse-grained description generated by iterating the one-step effective channel. Observers satisfying C1–C4 — the conditions that make their description non-Markovian, quantum for the source's finite-configuration class — cannot exist in the equilibrium phase of (S, φ). The reason is direct. C2 requires the written record to persist to readback — and in a thermal bath the record lives in correlations, so persistence needs the hidden sector's correlation time τ_B to exceed the observer's measurement timescale τ_S (the conserve-and-route alternative being unavailable in an equilibrated bath). In equilibrium, local correlations decay on thermal timescales (a picosecond or so at ordinary temperatures), for a generic thermal bath (protected slow modes excepted), while any physical observer's internal processing time is many orders of magnitude longer. So in equilibrium, C2 fails — and with it C4: a bath that thermalizes between steps erases the stored record before it can be read (the erasure lemma, [Main §2.3]). Since C4 is what the characterization requires, no observer satisfying the framework's definition can exist there.

This has two striking consequences. First, the "Boltzmann brain" problem dissolves at the level of definitions. A Boltzmann brain is by construction a local fluctuation out of a thermally equilibrated environment — that's what "fluctuation from equilibrium" means. Its hidden sector is therefore the thermally equilibrated bath around it, with τ_B bounded by thermal decorrelation — too short for C2 to hold, by six or more orders of magnitude for any brain-like observer when the bath's correlation time is set by the thermal scale (at most about 10^{-12} s at temperatures that support structured matter, against an internal processing time of at least about 10^{-6} s), a generic thermal estimate that fails for protected slow modes. In the coarse-grained description generated by iterating the one-step effective channel, Boltzmann brains carry no memory-bearing quantum phenomenology — no persistent record, no readback — though their statistics, like any finite law's, admit the representation; they aren't observers in the framework's sense. The exclusion is structural, not anthropic — it doesn't depend on any argument about "typical observers" or the trustworthiness of memory.

Second, the "past hypothesis" of cosmology gets replaced by a conditional theorem. Every standard cosmology has a mystery at t=0: why was the early universe so low-entropy? Boltzmann, Penrose, Albert, and Carroll all treat this as a postulate that has to be assumed separately. The framework replaces the postulate with a derivation. You don't need to posit low initial entropy to explain why we see it; you need the definition of observation together with the mixing hypothesis for equilibrium regions, which select for the non-equilibrium phase. Observers exist where C1–C4 hold. Under that hypothesis, and in the coarse-grained description generated by iterating the one-step effective channel, C1–C4 don't hold in equilibrium. Therefore observers exist in non-equilibrium. The mixing hypothesis is the premise that takes the place of the postulate.

In our universe, the non-equilibrium phase is provided by the expanding cosmological horizon. The horizon's expansion sets $\tau_B \sim H^{-1}$ — about 10^{17} seconds — which is structurally (not thermally) generated and is vastly longer than any laboratory timescale. This is why we see the memory-bearing quantum phenomenology at all: we're in the one phase of the universe where the framework's observer-conditions can be met — representability itself is universal; what the phase selects is the nontrivial sector. Entropy increase, the Second Law, and the felt direction of time emerge as coarse-grained descriptions of the dynamics accessible to observers in this non-equilibrium phase. Like quantum mechanics itself, the direction of time is emergent — a consequence of observing a lossless cycle from within, during the phase when such observation is structurally possible.

The incompleteness family

This places the framework in a family of results where self-reference under finite resources produces rigid structure.

Gödel (1931): A sufficiently powerful formal system cannot prove all truths about itself. The unprovable statements aren't random — they have a precise structure (the Gödel sentence, the consistency statement, the arithmetical hierarchy).

Turing (1936): A universal computer cannot decide all questions about its own behavior. The undecidable problems aren't random — they have a precise structure (the halting problem, Turing degrees, the oracle hierarchy).

OI: A specified observation interface determines which state distinctions are present in its retained records. Finite observable laws admit the exact stochastic/reversible/fixed-basis quantum correspondence. Neither that correspondence nor C4 says that every hidden state remains unidentified; full operational quantum mechanics and its Bell-compatible physical realization have their separate hypotheses.

For a known finite model, ask which initial states are compatible with the retained record. A quantity is inferred exactly when it is constant on that record fibre. The swap map $(v, h) \mapsto (h, v)$ has four distinct two-reading records, so it identifies the full initial state and the current hidden state. Its uniform visible process still has $X_2 = X_0$ and one bit of conditional readback memory. This separates predictive hidden memory from hidden information left unresolved by the whole history.

Model redundancy is a separate issue. An independently prepared, exactly decoupled ancillary sector leaves the chosen visible channel unchanged under the hypotheses of [Substratum Theorem 24(iii)]. Alphabet-size comparisons retain their stated additional hypotheses. These are comparisons of accessible laws across realizations, rather than a theorem that every state of one fixed realization is unknown. Non-identifiability in a known finite model is not the halting problem: the compatible sets are computably finite.

The connection with computation is a partial read/write interface to a reversible update. It is an analogy about access, not an identification of finite state inference with Turing undecidability. The proved correspondence concerns finite observable laws; its memory clause concerns histories with different next-step conditionals. Physical limits on storing records, choosing interventions and computing from them must be supplied where used.

Gödel and Turing are usually read as negative results — limits on knowledge. The framework recasts the physical instance as *generative*. What is mathematically rigid here is the finite stochastic/reversible/fixed-basis quantum representation equivalence, together with the memory-bearing sector selected by readback. Calling the resulting description **full operational quantum mechanics** requires the additional coherent/composite lifting and locality conditions stated above; the stronger uniqueness claim is not being smuggled into the analogy.

Why does mathematics describe physics?

This reframes one of the oldest questions in the philosophy of science.

The standard view: mathematics *describes* physics. The physical world exists independently, and we discover that mathematical equations capture its behavior. Wigner called this the "unreasonable effectiveness of mathematics" — why should abstract structures developed by pure mathematicians turn out to describe the physical world?

Tegmark's answer: the physical world *is* a mathematical structure. But this is a metaphysical claim that can't be tested, and it doesn't explain *which* mathematical structure or *why* this particular one.

The OI framework provides a different answer, with a mechanism. Mathematics doesn't describe physics. Physics is what mathematics *looks like from inside*.

The mathematical structure is (S, φ) — a finite set and a bijection. An observer embedded in that structure, seeing only part of it through the partition V, produces a stochastic description of what it can see, and that finite observable law has an exact fixed-basis quantum representation. Not because QM was designed to describe nature, and not because nature is mysteriously mathematical, but because the structural constraints of embedded observation determine this representation class. The step from that exact representation class to the unique full local operational theory is the explicitly separated lifting problem.

The trace-out is the mechanism. It takes the complete mathematical structure — the full evolution matrix with all its eigenvalues, Jordan blocks, and nilpotent monodromy — and projects it onto what the observer can see: the semisimple spectral data, organized by the representation theory of the partition. Physics is this projection.

What gets erased? The nilpotent part — the Jordan blocks that exist over finite fields but vanish over the reals, the monodromy operator that the observer can't detect, the arithmetic data about eigenvalue orders in finite field extensions. This is genuine mathematical structure. It's *there* in the dynamics. But the observer, looking through the keyhole of the partition, sees only the diagonalizable shadow.

This gives a structural interpretation of the use of mathematics in physics: a model and observer interface generate accessible laws, which admit the representation described above. Theorem 23 adds conditional local outputs of a selected cubic model, not a proof that observations recover one complete mathematical ontology. Which additional microscopic differences are inaccessible depends on the specified observations. A complete classification cannot be inferred from the interpretation alone.

The converse is an open problem. Starting from observations, E1–E7 and A1–A6 do not by themselves select the local cubic branch used in Theorem 23. Its

wave rule, carrier and physical hypotheses are additional inputs. A common Bell-compatible completion and its uniqueness impose further obligations.

The forward construction starts with a model, a preparation rule and an observer interface, then derives accessible data. An inverse reconstruction would start with those data and identify which physical models realize them. Conditional outputs within a chosen model do not complete that second task.

Finite reversible observer models can represent finite operational laws, but that universal realization does not uniquely select the physical substratum. On a *selected* cubic nearest-neighbor wave-rule branch with H-link, the single-copy condition and the other physical hypotheses of [SM], the local representation calculations give $K = 6$ and multiplicities $(3, 2, 1)$. The corresponding Standard Model matter conclusions retain H-cust, H-spin', H-χ' and their separate premises. The H-Bell compatibility of one such model with Bell-violating composites remains conditional; the GR gap equation fixes a relation among $\hbar$, ϵ and σ only under its separate horizon/infrared conditions. The observed inputs and A1–A6 do not yet uniquely select this branch.

Substratum §4 distinguishes three observational notions. For a *fixed coherent-channel interface*, a single hidden permutation independent of the visible state and an independent decoupled normalized hidden extension preserve the complete channel family, but the six-state Pair C witness shows these restricted moves are **not exhaustive**. For *population-only observations*, a hidden relabelling depending on the visible input fibre preserves all rooted transition counts, a broader invariance. Equivalence *across all observer partitions* is a separate unresolved classification requiring correspondence of preparations and readouts. Alphabet-size comparisons retain the unconditional q-insensitive results of [SM §2.7], with transition-law equivalence across q requiring an additional hypothesis; graph universality carries the specific Theorems 1–6 structural content but not blanket equality of channels. The valid spanning-class block-word theorem (Substratum Proposition 24.2) does not turn equality of single-time channels into hidden-conjugation uniqueness.

The analogy with coordinate and gauge redundancy is a useful interpretive goal, but in OI the physically correct observational carrier and the permitted equivalence must first be determined. Theorem 23 currently gives conditional local structural outputs on a selected branch; it does not prove uniqueness among all substrata, and H-Bell adds an independent completion question. The existence of genuinely observationally redundant hidden coordinates does not mean every microscopic distinction is gauge.

Whether a mathematical structure is called reality or a description does not change measurements if both readings stipulate the same accessible data. This is a conditional statement about those readings, not a proof that every proposed ontology has identical observations. Theorem 23 does not identify a unique physical branch, and Theorem 24 leaves the full observational classification open.

The comparison with electromagnetic gauge freedom is useful when the redun-

dancy is specified. In OI a hidden-only relabelling preserves coherent channels under the stated hypotheses, while a more general hidden relabelling can preserve populations and alter coherences. Such examples do not identify an exact boundary between every physical and unphysical microscopic distinction.

From an embedded observer's perspective, observable mathematics means the model structures represented in a specified preparation/readout interface. The selected branch gives conditional candidates for physical structure, but no theorem here divides every microscopic feature into a uniquely recovered physical residue and a complete gauge remainder. The choice of carrier matters, and comparisons of alphabet size or graph structure retain their own scopes.

Wigner's puzzle can thus be approached through a model of accessible mathematical structure. This is an interpretive proposal supported by particular constructions, not a theorem identifying all physical reality with one reconstructed quotient. The physical selector and the full observational classification remain research questions.

The comparison with Gödel and Turing concerns limits of access, proof and computation. It does not establish that embedded observation uniquely forces every microscopic feature of physics. The finite-law representation, the selected cubic calculation and the horizon construction retain their separate premises and evidential status.

This is a companion overview to the framework's five core papers: "The Incompleteness of Observation" (Maybaum, [Main]), which presents the foundational embedded-observation theorems with detailed derivations; "The Standard Model from a Cubic Lattice" (Maybaum, [SM]), which derives the Standard Model's gauge group, matter content, and chiral structure from lattice dynamics; "The Gravitational Sector from the Cosmological Horizon" (Maybaum, [GR]), which relates $\hbar$ to the discreteness scale, recovers the Bekenstein-Hawking area law, and develops dark-sector phenomenology; "The Substratum: Reconstruction Theorem and Gauge Group" (Maybaum, [Substratum]), which addresses the ontological status of the lattice and proves the framework's reconstruction theorem; and "Hierarchical Structural Realism and Universality Classes of Embedded Observers" (Maybaum, [Structure]), which articulates the framework's two-dimensional hierarchical structure and develops its relationship to other unification programs. The philosophical lineage section draws on a systematic analysis mapping the framework's claims against the major traditions in Western, Eastern, and contemporary philosophy of science.

Appendix: Mathematical Summary Tables

Sections 1–2: What Each Piece Does

Component	What it establishes	What it uses
Definition + Lemmas 1–3	The physical setup (no QM assumed)	Nothing — these are starting points
Partition-relativity (§1.4)	Under the canonical measure, emergent description depends only on partition	Definition, Lemmas 2, 3
Emergent stochasticity (§2.1)	Determinism looks random to embedded observer	Definition, Lemmas 2, 3
P-indivisibility proof (§2.3)	The stochastic process is P-indivisible across the recurrence cycle (not, by itself, on accessible windows)	Definition, Lemma 1, a non-permutation T
Accessible-timescale lemma (§2.3)	Non-Markovianity is observable, not just formal	C1–C3, plus C4 holding quantitatively: two histories with the same present whose next-step laws differ by $\delta > 0$ (then $\mathrm{I} \geq \mathrm{p}_0\delta^2/\ln 2$)
Coin-and-die model (§2.4)	Concrete demonstration of all mechanisms	Definition + all conditions
Stochastic-quantum correspondence (§3.1)	every process in the source's class embeds in a unistochastic one; P-indivisibility makes it nontrivial	Barandes [10,11] or Stinespring (Appendix A)
Necessity proof (§3.3)	Accessible non-Markovian memory requires C1/C3/C4 in every faithful deterministic realization; C2's physical-timescale necessity conditional on the mixing hypothesis	Per-condition necessity ([Main §3.4])

Sections 3–6: The Logical Flow

Step	What it establishes	Key equation
§3.1: Stochastic-quantum correspondence	compressed ($\leq n^3$) representation for the correspondence's class; existence is internal and universal ($S \iff D \iff Q_{\mathrm{fb}}$, [Main §3.4])	T_ij = ancilla-marginal Born rule (T = $\lvert U\rvert^2$ in the trivial-ancilla case)
§3.1: Phase-locking	Under H-BG and the stated genericity hypotheses, Hamiltonian fixed up to shift, rephasing, and the antiunitary conjugate	Conditional classification and Fourier analysis
§3.2: Bell analysis	Deterministic screened graph-local completion is Bell-local; quantum CHSH therefore requires ontic parameter dependence with operational no-signaling	local deterministic completion $\Rightarrow S_{\mathrm{CHSH}} \leq 2$; stochastic causal-local class has the separate Tsirelson bound; bare/global OI has PR negative control
§3.3: Necessity of C1	Accessible memory requires coupling	No coupling → permutation → P-divisible
§3.3: Necessity of C2	Physical memory requires record persistence — slow-bath form conditional on mixing	Fast-mixing bath → Markov → P-divisible
§3.3: Necessity of C3	Accessible memory requires capacity	I $\leq \log_2$ m bounds memory
§3.4: Characterization theorem	non-Markovianity $\iff$ C1, C3, C4 per accessible horizon (C2 separate, conditional); representation equivalence $S \iff D \iff Q_{\mathrm{fb}}$ — internal, universal; universal hidden-memory theorem	Biconditional + internal permutation unitary (compressed form imported under (T))
§4: Cosmological horizon	The universe satisfies the definition	C1 ✓, C2 ✓, C3 ✓, C4 ✓

Step	What it establishes	Key equation
§5.1: Classical temperature	T_cl = $c^2\varepsilon^2\kappa/(8\pi$Gk_B$\sigma)$	No $\hbar$
§5.2 Step 1: Uniqueness	$\hbar$ is determined, not free	Partition-relativity
§5.2 Step 2: Boundary-only	$\hbar$ depends only on c, G, ε	Deep sector frozen
§5.2 Step 3: Dimensional analysis	$\hbar = \beta\ c^3\varepsilon^2/$G	Unique combination
§5.2 Step 4: Thermal matching	T_cl = T_Q fixes $\beta = 1/(4\sigma)$	$\hbar = c^3\varepsilon^2/(4G\sigma)$
§5.3: Bohr-frequency completeness	Frequencies determined, $\hbar$ not	Dimensional obstruction
§6: Self-consistency	$\varepsilon = 2\sqrt{\sigma}$ l_p, S_BH = A/(4l_p^2)	$\varepsilon^2 = 4\sigma\hbar$G$/c^3$

The chain from definition to $\hbar$:

Definition $\rightarrow$ universal Q_{fb} representation (conditional memory marking the memory sector) $\rightarrow$ Hamiltonian with unknown $\hbar \rightarrow$ thermal matching at the horizon $\rightarrow \hbar = c^3\varepsilon^2/(4G\sigma) \rightarrow \varepsilon = 2\sqrt{\sigma}$ l_p $\rightarrow$ S_BH = A/(4 l_p^2)

Every link is either a proof or a calculation. No link requires quantum mechanics as input — QM appears as output at step 3 and its parameters are determined by steps 4–6.

Appendix: Key Mathematical Subtleties

1. The gap equation is a gap equation, not a derivation from nothing.

The framework has one free geometric parameter (ε). The gap equation relates ε to $\hbar$ at given σ. You need to know one to determine the other. The specific relationship $\hbar = c^3\varepsilon^2/(4G\sigma)$ enters three later results: the Bekenstein-Hawking formula with the factor 1/4 (§6), the compression ratio S_dS of the conditional CC dissolution (§7.3), and the Type II running-vacuum reading of the dark energy (§8.1). The three relations share the calibration; they are consequences of it, not independent checks of it. The situation is analogous to deriving the Schwarzschild metric with M as a free parameter: the derivation has genuine content (the functional form) even though one input is not determined from within.

2. The KMS condition on the lattice.

The thermal matching in Step 4 uses the KMS condition, which is proved for continuum QFT on curved spacetime. The emergent QFT from Part I is lattice-regularized with cutoff ε. Thermal periodicity on a lattice has corrections of

order $(\varepsilon\kappa/c^2)^2$. For the cosmological horizon at $\sigma = 1$: $\varepsilon\kappa/c^2 = \varepsilon H/c \sim 10^{-61}$, so corrections are $\sim 10^{-122}$. Negligible, but a lattice-native proof of the KMS condition would strengthen the argument.

3. The D-gauge reconstruction requires genericity and is a conditional theorem.

The phase-locking argument assumes non-degenerate spectrum, non-degenerate energy gaps, and non-vanishing overlaps (conditions G1–G3); the two-branch reconstruction is a conditional theorem of [GR §3.3]: for two Hamiltonians on the same finite configuration space of dimension at least three, each with distinct nonzero energy gaps and non-vanishing configuration-basis overlaps, equal transition probabilities imply $H' = DHD^\dagger + E_0$ or $H' = -DHD^\dagger + E_0$, for a time-independent diagonal unitary D and a real E_0, conditional on H-BG — every non-congruent pair of strictly ordered integer rulers with distinct positive differences and equal difference sets has six marks and, after relabeling, realizes either μ or its global reversal, where μ sends each ascending mark pair of $\{0,1,4,10,12,17\}$ to the ascending mark pair of $\{0,1,8,11,13,17\}$ with the same difference, the orientation chosen once for the whole pair. The complete comparison of H-BG with the published integer classification of Bekir and Golomb is open; Bohr-frequency completeness itself needs only the spectrum and overlap conditions. These fail on a measure-zero set of Hamiltonians. For the cosmological realization, a stronger argument applies to G1 specifically: the genericity conditions concern the *effective* visible-sector Hamiltonian $\hat{H}_{\text{eff}}$ (after the trace-out), not the total Hamiltonian H_{tot}. The relevant distinction is whether a symmetry of H_{tot} preserves the visible/hidden factorization. The spatial symmetries an observer-centered partition breaks — translation, rotation about distant points, boost — are *factorization-breaking* (they map boundary visible sites to hidden sites across the cosmological horizon), so they have no action on the visible factor alone and the degeneracies they protect are lifted in $\hat{H}_{\text{eff}}$. Factorization-preserving symmetries $S_V \otimes S_H$, by contrast, descend to a visible symmetry S_V of $\hat{H}_{\text{eff}}$ (the uniform-bath trace-out is invariant under any unitary on the hidden factor); the visible-internal ones among these are the gauge symmetries of the emergent QFT, which correspond to physically equivalent states and don't affect the phase-locking argument. G2 (non-degenerate gaps) is not symmetry-protected and rests on genericity directly; it holds for generic $\hat{H}_{\text{eff}}$ and can fail only on a measure-zero set, with no known structural feature of the substratum forcing systematic failure (single-cycle hidden dynamics in particular does not, since the Hermitian $\hat{H}_{\text{eff}}$ — unlike the unitary cycle operator's equally-spaced phases — has unequally-spaced real eigenvalues and retains non-degenerate gaps after the trace-out).

4. Why doesn't the deep sector matter?

The boundary-only dependence lemma is crucial because without it, $\hbar$ might depend on the vast interior of the hidden sector — which we can't observe or constrain. The lemma shows the deep sector is frozen on observable timescales,

so its details are irrelevant. $\hbar$ is a property of the boundary geometry, not of the bulk.

Acknowledgment of AI-Assisted Technologies: The author acknowledges the use of **Claude Opus 4.6** and **Gemini 3.1 Pro** to assist in synthesizing technical concepts and refining clarity. The final text and all scientific claims were reviewed and verified by the author.